

LECTURE NOTES

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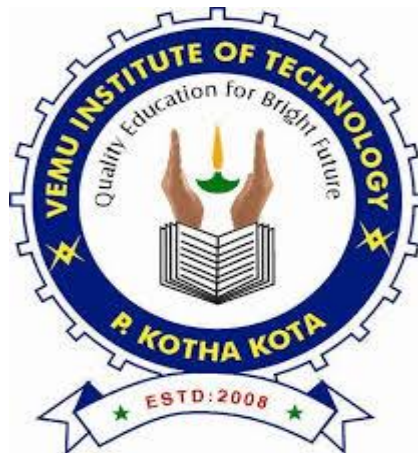
ELECTRICAL MACHINES - II

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UNIT-I

Single Phase Transformers

Introduction

The transformer is a device that transfers electrical energy from one electrical circuit to another electrical circuit. The two circuits may be operating at different voltage levels but always work at the same frequency. Basically transformer is an electro-magnetic energy conversion device. It is commonly used in electrical power system and distribution systems. It can change the magnitude of alternating voltage or current from one value to another. This useful property of transformer is mainly responsible for the widespread use of alternating currents rather than direct currents i.e., electric power is generated, transmitted and distributed in the form of alternating current. Transformers have no moving parts, rugged and durable in construction, thus requiring very little attention. They also have a very high efficiency as high as 99%.

Single Phase Transformer

A transformer is a static device of equipment used either for raising or lowering the voltage of an a.c. supply with a corresponding decrease or increase in current. It essentially consists of two windings, the primary and secondary, wound on a common laminated magnetic core as shown in Fig 1. The winding connected to the a.c. source is called primary winding (or primary) and the one connected to load is called secondary winding (or secondary). The alternating voltage V_1 whose magnitude is to be changed is applied to the primary.

Depending upon the number of turns of the primary (N_1) and secondary (N_2), an alternating e.m.f. E_2 is induced in the secondary. This induced e.m.f. E_2 in the secondary causes a secondary current I_2 . Consequently, terminal voltage V_2 will appear across the load.

If $V_2 > V_1$, it is called a step up-transformer.

If $V_2 < V_1$, it is called a step-down transformer.

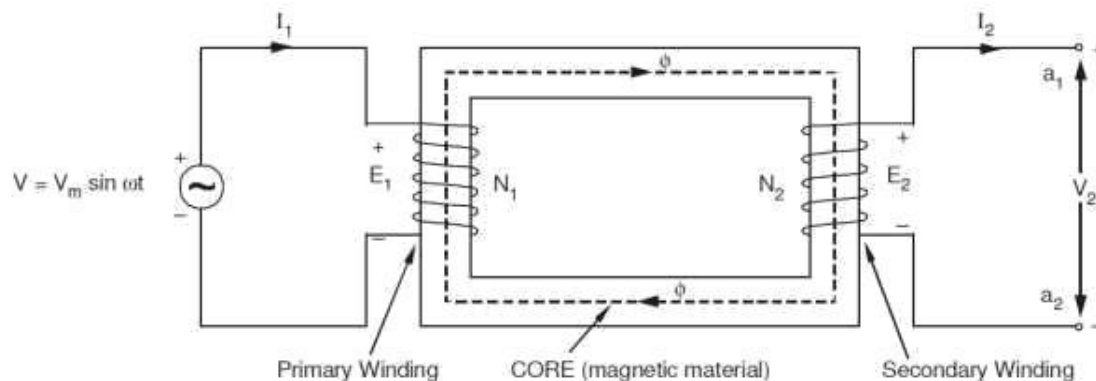


Fig. 2.1 Schematic diagram of single phase transformer

Constructional Details

Depending upon the manner in which the primary and secondary windings are placed on the core, and the shape of the core, there are two types of transformers, called (a) core type, and (b) shell type.

Core-type and Shell-type Construction

In core type transformers, the windings are placed in the form of concentric cylindrical coils placed around the vertical limbs of the core. The low-voltage (LV) as well as the high-voltage (HV) winding are made in two halves, and placed on the two limbs of core. The LV winding is placed next to the core for economy in insulation cost. Figure 2.1(a) shows the cross-section of the arrangement. In the shell type transformer, the primary and secondary windings are wound over the central limb of a three-limb core as shown in Figure 2.1(b). The HV and LV windings are split into a number of sections, and the sections are interleaved or sandwiched i.e. the sections of the HV and LV windings are placed alternately.

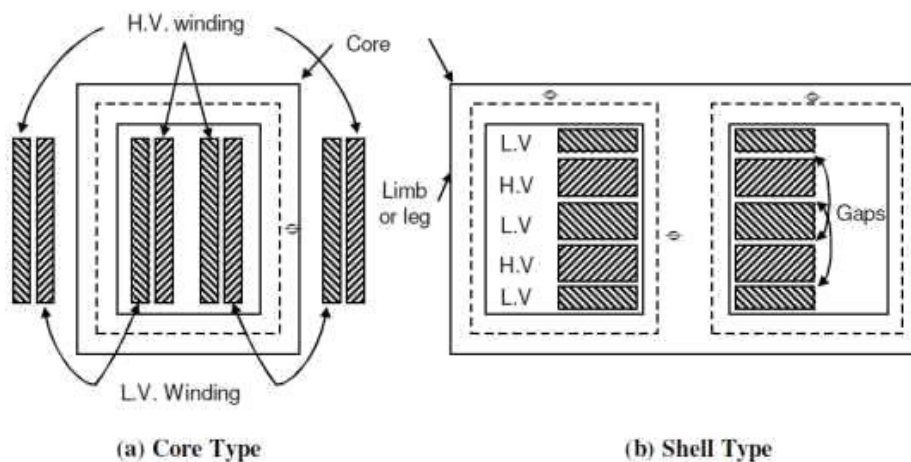


Fig: 2.1 Core type & shell type transformer

Core

The core is built-up of thin steel laminations insulated from each other. This helps in reducing the eddy current losses in the core, and also helps in construction of the transformer. The steel used for core is of high silicon content, sometimes heat treated to produce a high permeability and low hysteresis loss. The material commonly used for core is CRGO (Cold Rolled Grain Oriented) steel. Conductor material used for windings is mostly copper. However, for small distribution transformer aluminum is also sometimes used. The conductors, core and whole windings are insulated using various insulating materials depending upon the voltage.

Insulating Oil

In oil-immersed transformer, the iron core together with windings is immersed in insulating oil. The insulating oil provides better insulation, protects insulation from moisture and transfers the heat produced in core and windings to the atmosphere. The transformer oil should possess the following qualities:

- (a) High dielectric strength,
- (b) Low viscosity and high purity,
- (c) High flash point, and
- (d) Free from sludge.

Transformer oil is generally a mineral oil obtained by fractional distillation of crude oil.

Tank and Conservator

The transformer tank contains core wound with windings and the insulating oil. In large transformers small expansion tank is also connected with main tank is known as conservator. Conservator provides space when insulating oil expands due to heating. The transformer tank is provided with tubes on the outside, to permits circulation of oil, which aides in cooling. Some additional devices like breather and Buchholz relay are connected with main tank. Buchholz relay is placed between main tank and conservator. It protect the transformer under extreme heating of transformer winding. Breather protects the insulating oil from moisture when the cool transformer sucks air inside. The silica gel filled breather absorbs moisture when air enters the tank. Some other necessary parts are connected with main tank like, Bushings, Cable Boxes, Temperature gauge, Oil gauge, Tapings, etc.

principle of Operation

When an alternating voltage V_1 is applied to the primary, an alternating flux ϕ is set up in the core. This alternating flux links both the windings and induces e.m.f.s E_1 and E_2 in them according to

Faraday's laws of electromagnetic induction. The e.m.f. E_1 is termed as primary e.m.f. and e.m.f. E_2 is termed as secondary e.m.f.

$$\begin{aligned}\text{Clearly, } E_1 &= -N_1 \frac{d\phi}{dt} \\ \text{and } E_2 &= -N_2 \frac{d\phi}{dt} \\ \therefore \frac{E_2}{E_1} &= \frac{N_2}{N_1}\end{aligned}$$

Note that magnitudes of E_2 and E_1 depend upon the number of turns on the secondary and primary respectively.

If $N_2 > N_1$, then $E_2 > E_1$ (or $V_2 > V_1$) and we get a step-up transformer. If $N_2 < N_1$, then $E_2 < E_1$

(or $V_2 < V_1$) and we get a step-down transformer.

If load is connected across the secondary winding, the secondary e.m.f. E_2 will cause a current I_2 to flow through the load. Thus, a transformer enables us to transfer a.c. power from one circuit to another with a change in voltage level.

The following points may be noted carefully

- (a) The transformer action is based on the laws of electromagnetic induction.
- (b) There is no electrical connection between the primary and secondary.
- (c) The a.c. power is transferred from primary to secondary through magnetic flux.

There is no change in frequency i.e., output power has the same frequency as the input power.

(e) The losses that occur in a transformer are:

(a) **core losses**—eddy current and hysteresis losses

(b) **copper losses**—in the resistance of the windings

In practice, these losses are very small so that output power is nearly equal to the input primary power. In other words, a transformer has very high efficiency.

E.M.F. Equation of a Transformer

Consider that an alternating voltage V_1 of frequency f is applied to the primary as shown in Fig.2.3. The sinusoidal flux ϕ produced by the primary can be represented as:

$$\phi = \phi_m \sin \omega t$$

When the primary winding is excited by an alternating voltage V_1 , it is circulating alternating current, producing an alternating flux ϕ .

ϕ - Flux

ϕ_m - maximum value of flux

N_1 - Number of primary turns

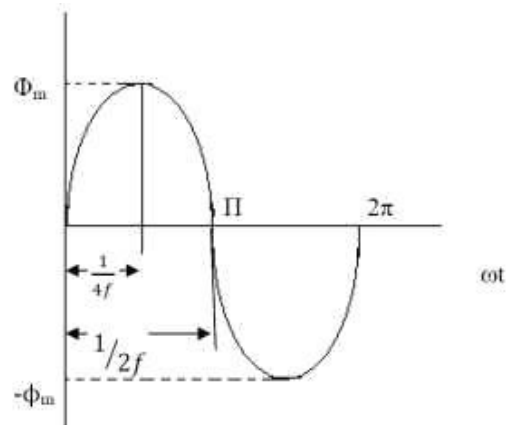
N_2 - Number of secondary turns

f - Frequency of the supply voltage

E_1 - R.M.S. value of the primary induced e.m.f

E_2 - R.M.S. value of the secondary induced e.m.f

The instantaneous e.m.f. e_1 induced in the primary is -



From Faraday's law of electromagnetic induction -

$$\text{Average e.m.f per turns} = \frac{d\phi}{dt}$$

$d\phi$ = change in flux

dt = time required for change in flux

The flux increases from zero value to maximum value ϕ_m in $1/4f$ of the time period that is in $1/4f$ seconds.

The change of flux that takes place in $1/4f$ seconds = $\phi_m - 0 = \phi_m$

webers **Voltage Ratio**

$$\frac{d\phi}{dt} = \frac{\phi_m}{1/4f} = 4f\phi_m \text{ w_b/sec.}$$

Since flux ϕ varies sinusoidally, the R.m.s value of the induced e.m.f is obtained by multiplying the average value with the form factor

$$\text{Form factor of a sinwave} = \frac{\text{R.m.s value}}{\text{Average value}} = 1.11$$

R.M.S Value of e.m.f induced in one turns = $4\phi_m f \times 1.11$ Volts.

$$= 4.44\phi_m f \text{ Volts.}$$

R.M.S Value of e.m.f induced in primary winding = $4.44\phi_m f N_1$ Volts.

R.M.S Value of e.m.f induced in secondary winding = $4.44\phi_m f N_2$ Volts.

The expression of E_1 and E_2 are called e.m.f equation of a transformer

$$\begin{aligned} V_1 &= E_1 = 4.44\phi_m f N_1 \text{ Volts.} \\ V_2 &= E_2 = 4.44\phi_m f N_2 \text{ Volts.} \end{aligned}$$

Voltage transformation ratio is the ratio of e.m.f induced in the secondary winding to the e.m.f induced in the primary winding.

$$\frac{E_2}{E_1} = \frac{4.44\phi_m f N_2}{4.44\phi_m f N_1}$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

This ratio of secondary induced e.m.f to primary induced e.m.f is known as voltage transformation ratio

$$E_2 = KE_1 \quad \text{where } K = \frac{N_2}{N_1}$$

1. If $N_2 > N_1$ i.e. $K > 1$ we get $E_2 > E_1$ then the transformer is called step up transformer.
2. If $N_2 < N_1$ i.e. $K < 1$ we get $E_2 < E_1$ then the transformer is called step down transformer.
3. If $N_2 = N_1$ i.e. $K = 1$ we get $E_2 = E_1$ then the transformer is called isolation transformer or 1:1 Transformer

Current Ratio

Current ratio is the ratio of current flow through the primary winding (I_1) to the current flowing through the secondary winding (I_2). In an ideal transformer -

Apparent input power = Apparent output power.

$$V_1 I_1 = V_2 I_2$$

$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

Volt-Ampere Rating

- i) The transformer rating is specified as the products of voltage and current (VA rating).
- ii) On both sides, primary and secondary VA rating remains same. This rating is generally expressed in KVA (Kilo Volts Amperes rating)

$$\frac{V_1}{V_2} = \frac{I_2}{I_1} = K$$

$$V_1 I_1 = V_2 I_2$$

$$\text{KVA Rating of a transformer} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000} \quad (1000 \text{ is to convert KVA to VA})$$

V_1 and V_2 are the V_r of primary and secondary by using KVA rating we can calculate I_1 and I_2 Full load current and it is safe maximum current.

$$I_1 \text{ Full load current} = \frac{\text{KVA Rating} \times 1000}{V_1}$$

$$I_2 \text{ Full load current} = \frac{\text{KVA Rating} \times 1000}{V_2}$$

Transformer on No-load

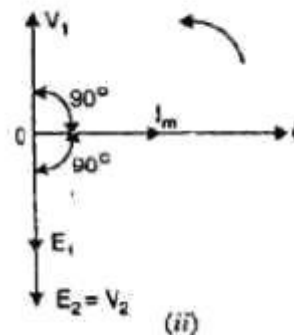
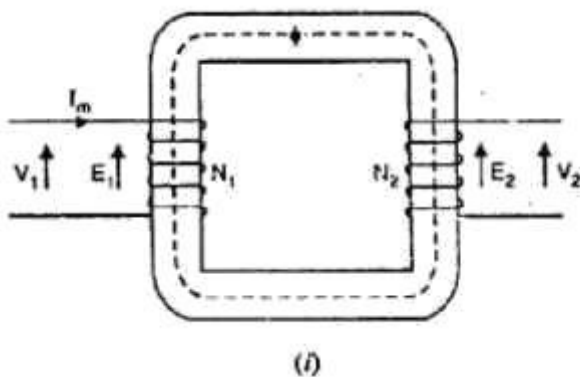
- a) Ideal transformer
- b) Practical transformer

a) Ideal Transformer

An ideal transformer is one that has

- (i) No winding resistance
- (ii) No leakage flux i.e., the same flux links both the windings
- (iii) No iron losses (i.e., eddy current and hysteresis losses) in the core

Although ideal transformer cannot be physically realized, yet its study provides a very powerful tool in the analysis of a practical transformer. In fact, practical transformers have properties that approach very close to an ideal transformer.



Consider an ideal transformer on no load i.e., secondary is open-circuited as shown in Fig.2.4 (i). under such conditions, the primary is simply a coil of pure inductance. When an alternating voltage V_1 is applied to the primary, it draws a small magnetizing current I_m which lags behind the applied voltage by 90° . This alternating current I_m produces an alternating flux ϕ which is proportional to and in phase with it. The alternating flux ϕ links both the windings and induces e.m.f. E_1 in the primary and e.m.f. E_2 in the secondary. The primary e.m.f. E_1 is, at every instant, equal to and in opposition to V_1 (Lenz's law). Both e.m.f.s E_1 and E_2 lag behind flux ϕ by 90° . However, their magnitudes depend upon the number of primary and secondary turns. Fig. 2.4 (ii) shows the phasor diagram of an ideal transformer on no load. Since flux ϕ is common to both the windings, it has been taken as the reference phasor. The primary e.m.f. E_1 and secondary e.m.f. E_2 lag behind the flux ϕ by 90° . Note that E_1 and E_2 are in phase. But E_1 is equal to V_1 and 180° out of phase with it.

$$\frac{E_2}{E_1} = \frac{V_2}{V_1} = K$$

Phasor Diagram

- i) Φ (flux) is reference
- ii) I_m produce ϕ and it is in phase with ϕ , V_1 Leads I_m by 90°

E_1 and E_2 are in phase and both opposing supply voltage V_1 , winding is purely inductive
So current has to lag voltage by 90° .

- iii) The power input to the transformer

$$P = V_1 I_1 \cos(90^\circ) \dots\dots\dots (\cos 90^\circ = 0)$$

$$P = 0 \text{ (ideal transformer)}$$

b)i) Practical Transformer on no load

A practical transformer differs from the ideal transformer in many respects. The practical

transformer has (i) iron losses (ii) winding resistances and (iii) Magnetic leakage

(i) Iron losses. Since the iron core is subjected to alternating flux, there occurs eddy current and hysteresis loss in it. These two losses together are known as iron losses or core losses. The iron losses depend upon the supply frequency, maximum flux density in the core, volume of the core etc. It may be noted that magnitude of iron losses is quite small in a practical transformer.

(ii) Winding resistances. Since the windings consist of copper conductors, it immediately follows that both primary and secondary will have winding resistance. The primary resistance R_1 and secondary resistance R_2 act in series with the respective windings as shown in Fig. When current flows through the windings, there will be power loss as well as a loss in voltage due to IR drop. This will affect the power factor and E_1 will be less than V_1 while V_2 will be less than E_2 .

Consider a practical transformer on no load i.e., secondary on open-circuit as Shown in Fig 2.5.

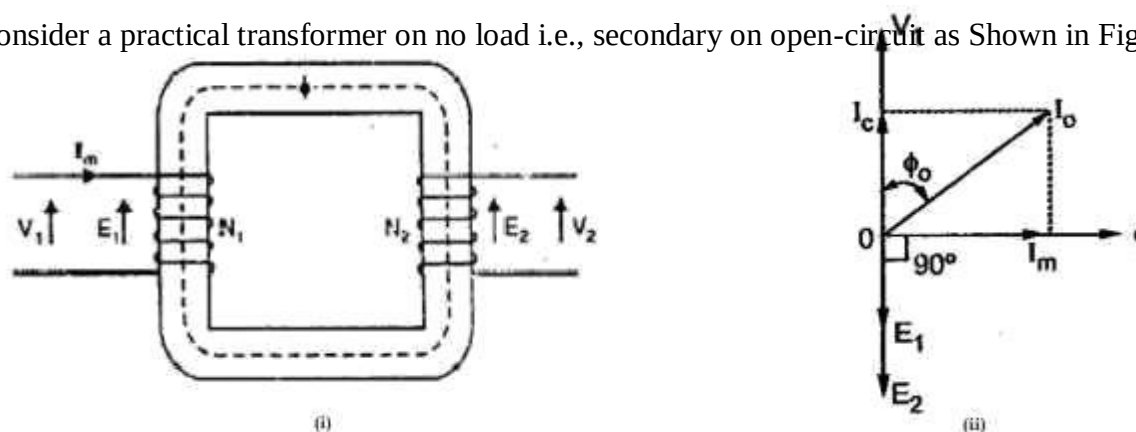


Fig: 2.5 Phasor diagram of transformer at no load

Here the primary will draw a small current I_0 to supply -

- (i) The iron losses and
- (ii) A very small amount of copper loss in the primary.

Hence the primary no load current I_0 is not 90° behind the applied voltage V_1 but lags it by an angle $\phi_0 < 90^\circ$ as shown in the phasor diagram. No load input power, $W_0 = V_1 I_0 \cos \phi_0$

As seen from the phasor diagram in Fig.2.5 (ii), the no-load primary current I_0

(i) the component I_c in phase with the applied voltage V_1 . This is known as active or working or iron loss component and supplies the iron loss and a very small primary copper loss.

$$I_c = I_0 \cos \phi_0$$

The component I_m lagging behind V_1 by 90° and is known as magnetizing component. It is this component which produces the mutual flux ϕ in the core.

$$I_m = I_0 \sin \phi_0$$

Clearly, I_0 is phasor sum of I_m and I_c ,

$$I_0 = \sqrt{I_m^2 + I_c^2}$$

$$\text{No load P.F., } \cos \phi_0 = \frac{I_c}{I_0}$$

The no load primary copper loss (i.e. $I_0^2 R_1$) is very small and may be neglected. Therefore, the no load primary input power is practically equal to the iron loss in the transformer i.e., No load input power, $W_0 = V_1 I_0 \cos \phi_0 = P_i = \text{Iron loss}$

b) ii) Practical Transformer on Load

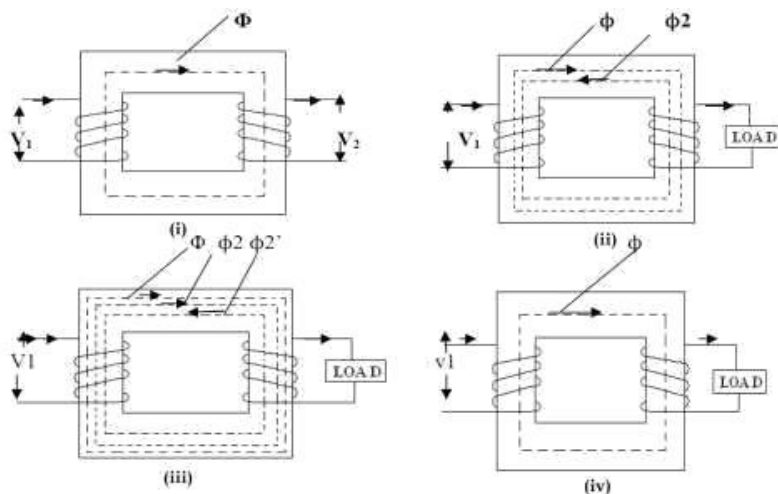


Fig: 2.6

At no load, there is no current in the secondary so that $V_2 = E_2$. On the primary side, the drops in R_1 and X_1 , due to I_0 are also very small because of the smallness of I_0 . Hence, we can say that at no load, $V_1 = E_1$.

i) When transformer is loaded, the secondary current I_2 flows through the secondary winding.

- ii) Already I_m magnetizing current flow in the primary winding fig. 2.6(i).
- iii) The magnitude and phase of I_2 with respect to V_2 is determined by the characteristics of the load.
 - a) I_2 in phase with V_2 (resistive load)
 - b) I_2 lags with V_2 (Inductive load)
 - c) I_2 leads with V_2 (capacitive load)
- iv) Flow of secondary current I_2 produce new Flux ϕ_2 fig.2.6 (ii)
- v) Φ is main flux which is produced by the primary to maintain the transformer as constant magnetising component.
- vi) Φ_2 opposes the main flux ϕ , the total flux in the core reduced. It is called demagnetising Ampere- turns due to this E_1 reduced.
- vii) To maintain the ϕ constant primary winding draws more current (I_2') from the supply (load component of primary) and produce ϕ_2' flux which is oppose ϕ_2 (but in same direction as ϕ), to maintain flux constant in the core fig.2.6 (iii).
- viii) The load component current I_2' always neutralizes the changes in the load.
- ix) Whatever the load conditions, the net flux passing through the core is approximately the same as at no-load. An important deduction is that due to the constancy of core flux at all loads, the core loss is also practically the same under all load conditions fig.2.6 (iv).

$$\Phi_2 = \phi_2', \quad N_2 I_2 = N_1 I_2', \quad I_2' = \frac{N_2}{N_1} X I_2 = K I_2$$

Phasor Diagram

- i) Take (ϕ) flux as reference for all load
- ii) The no load I_0 which lags by an angle ϕ_0 . $I_0 = \sqrt{I_c^2 + I_m^2}$
- iii) The load component I_2' , which is in anti-phase with I_2 and phase of I_2 is decided by the load.
- iv) Primary current I_1 is vector sum of I_0 and I_2'

$$\vec{I}_1 = \vec{I}_0 + \vec{I}_2'$$

$$I_1 = \sqrt{I_0^2 + I_2'^2}$$

- a) If load is Inductive, I_2 lags E_2 by ϕ_2 , shown in phasor diagram fig 2.7 (a).
- b) If load is resistive, I_2 in phase with E_2 shown in phasor diagram fig. 2.7 (b).
- c) If load is capacitive load, I_2 leads E_2 by ϕ_2 shown in phasor diagram fig. 2.7 (c).

For easy understanding at this stage here we assumed E_2 is equal to V_2 neglecting various drops.

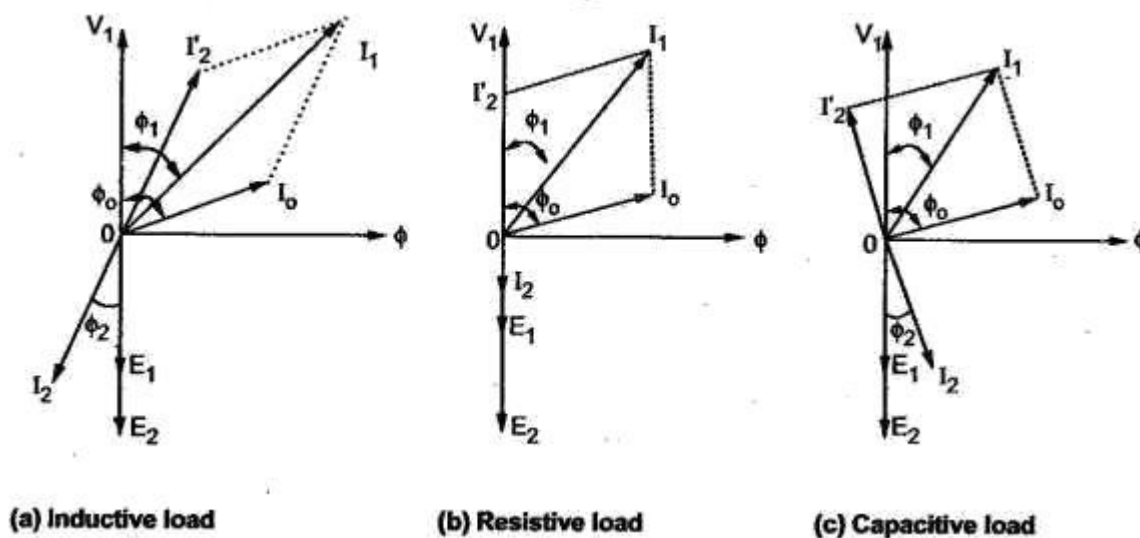


Fig: 2.7.a

$$\vec{I_1} = \vec{I_0} + \vec{I_2}$$

$$I_1 \cong I_2'$$

Balancing the ampere – turns

$$I_1 = \sqrt{I_0^2 + I_2'^2}$$

$$N_1 I_1' = N_1 I_1 + N_2 I_2$$

$$\frac{I_1}{I_2} = \frac{N_2}{N_1} = K$$

Now we going to construct complete phasor diagram of a transformer (shown in Fig: 2.7.b)

Effect of Winding Resistance

In practical transformer it process its own winding resistance causes power loss and also the voltage drop.

R1 – primary winding resistance in ohms.

R2 – secondary winding resistance in ohms.

The current flow in primary winding make voltage drop across it is denoted as $I_1 R_1$ here supply voltage V_1 has to supply this drop primary induced e.m.f E_1 is the vector difference between V_1 and $I_1 R_1$.

$$\vec{E}_1 = \vec{V}_1 - \vec{I}_1 R_1$$

Similarly the induced e.m.f in secondary E_2 , The flow of current in secondary winding makes voltage drop across it and it is denoted as $I_2 R_2$ here E_2 has to supply this drop. The vector difference between E_2 and $I_2 R_2$

$$\vec{V}_2 = \vec{E}_2 - \vec{I}_2 R_2 \quad (\text{Assuming as purely resistive drop here})$$

Equivalent Resistance

- 1) It would now be shown that the resistances of the two windings can be transferred to any one of the two winding.
- 2) The advantage of concentrating both the resistances in one winding is that it makes calculations very simple and easy because one has then to work in one winding only.
- 3) Transfer to any one side either primary or secondary without affecting the performance of the transformer.

The total copper loss due to both the resistances

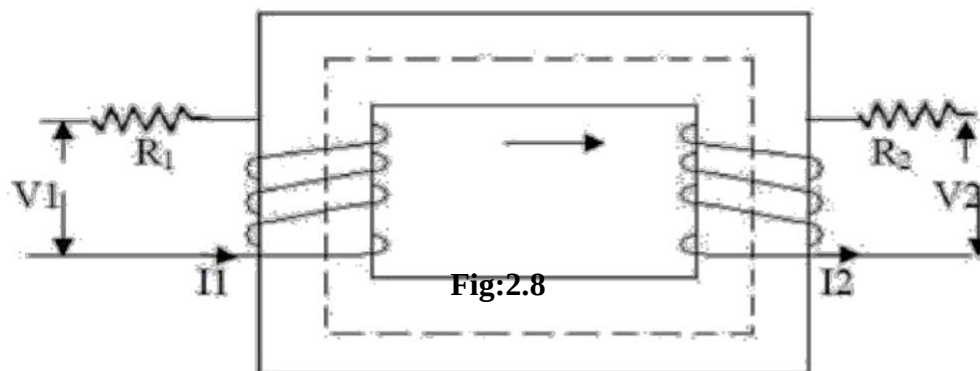
$$\begin{aligned} \text{Total copper loss} &= I_1^2 R_1 + I_2^2 R_2 \\ &= I_1^2 \left[R_1 + \frac{I_2^2}{I_1^2} R_2 \right] \\ &= I_1^2 \left[R_1 + \frac{1}{K} R_2 \right] \end{aligned}$$

$\frac{R_2}{K^2}$ is the resistance value of R_2 shifted to primary side and denoted as R_2' .
 R_2' is the equivalent resistance of secondary referred to primary

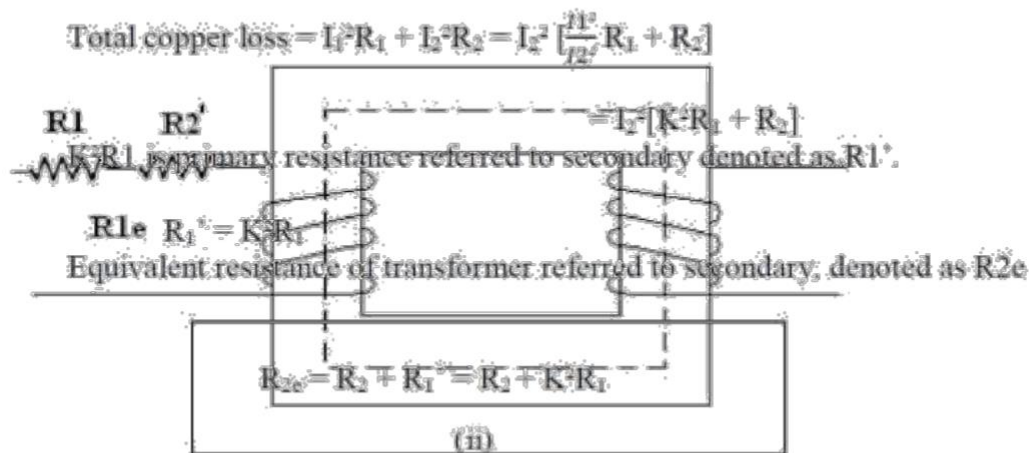
$$R_2' = \frac{R_2}{K^2}$$

Equivalent resistance of transformer referred to primary fig (ii)

$$R_{1e} = R_1 + R_2' = R_1 + \frac{R_2}{K^2}$$



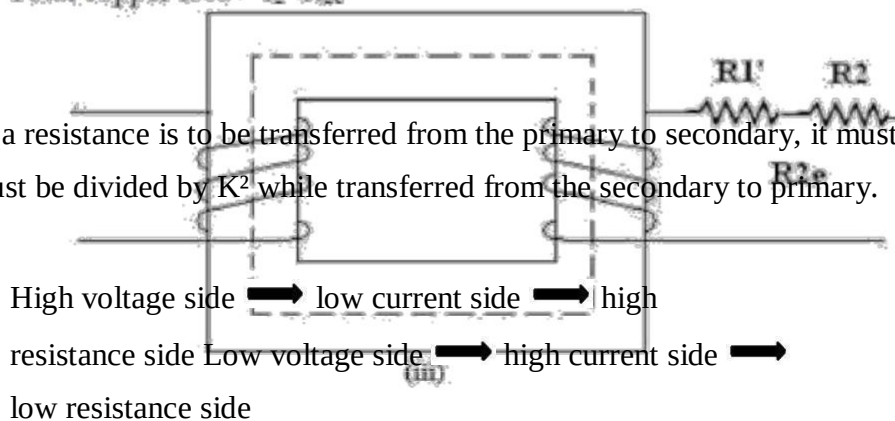
Similarly it is possible to refer the equivalent resistance to secondary winding.



Total copper loss = $I_2^2 R_{2e}$

Note:

- i) When a resistance is to be transferred from the primary to secondary, it must be multiplied by K^2 , it must be divided by K^2 while transferred from the secondary to primary.



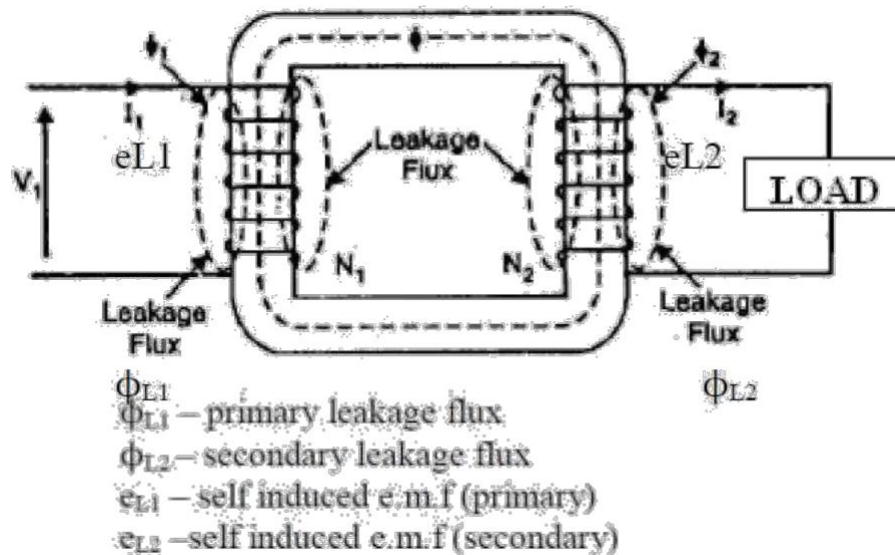
Effect of Leakage Reactance

- i) It has been assumed that all the flux linked with primary winding also links the secondary winding. But, in practice, it is impossible to realize this condition.

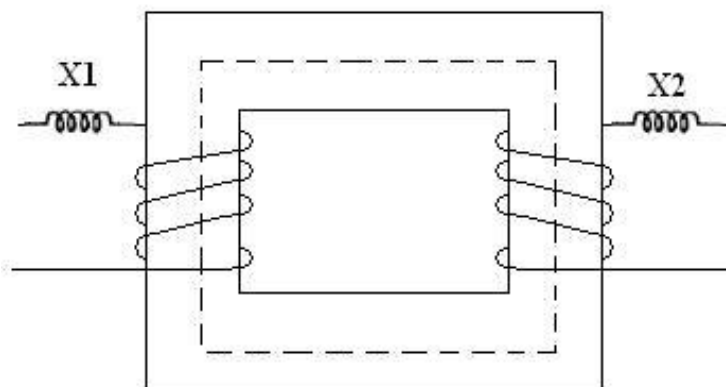
ii) However, primary current would produce flux ϕ which would not link the secondary winding. Similarly, current would produce some flux ϕ that would not link the primary winding.

iii) The flux ϕ_{L1} complete its magnetic circuit by passing through air rather than around the core, as shown in fig.2.9. This flux is known as primary leakage flux and is proportional to the primary ampere – turns alone because the secondary turns do not links the magnetic circuit of ϕ_{L1} . It induces an e.m.f e_{L1} in primary but not in secondary.

iv) The flux ϕ_{L2} complete its magnetic circuit by passing through air rather than around the core, as shown in fig. This flux is known as secondary leakage flux and is proportional to the secondary ampere– turns alone because the primary turns do not links the magnetic circuit of ϕ_{L2} . It induces an e.m.f e_{L2} in secondary but not in primary.



Equivalent Leakage Reactance



Similarly to the resistance, the leakage reactance also can be transferred from primary to secondary. The relation through K^2 remains same for the transfer of reactance as it is studied earlier for the resistance

X_1 – leakage reactance of primary.

X_2 - leakage reactance of secondary.

Then the total leakage reactance referred to primary is X_{1e} given by

$$X_{1e} = X_1 + X_2'$$

$$X_2' = \frac{X_2}{K^2}$$

The total leakage reactance referred to secondary is X_{2e} given by

$$X_{2e} = X_2 + X_1'$$

$$X_1' = K^2 X_1$$

$$\begin{aligned} X_{1e} &= X_1 + X_2' \\ X_{2e} &= X_2 + X_1' \end{aligned}$$

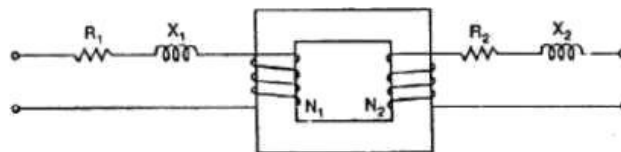
Equivalent Impedance

The transformer winding has both resistance and reactance (R_1, R_2, X_1, X_2). Thus we can say that the total impedance of primary winding is Z_1 which is,

$$Z_1 = R_1 + jX_1 \text{ ohms}$$

On secondary winding,

$$Z_2 = R_2 + jX_2 \text{ ohms}$$

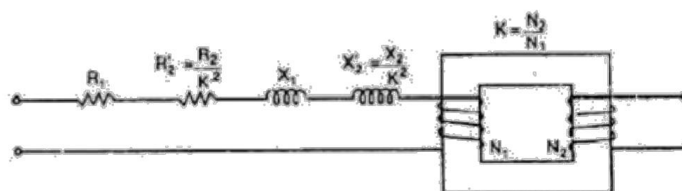


Individual magnitude of Z_1 and Z_2 are

$$Z_1 = \sqrt{R_1^2 + X_1^2}$$

$$Z_2 = \sqrt{R_2^2 + X_2^2}$$

Similar to resistance and reactance, the impedance also can be referred to any one side,



Z_{1e} = total equivalent impedance referred to primary

$$Z_{1e} = R_{1e} + jX_{1e} = Z_1 + Z_2' = Z_1 + \frac{Z_2}{K^2}$$

Complete Phasor Diagram of a Transformer (for Inductive Load or Lagging pf)

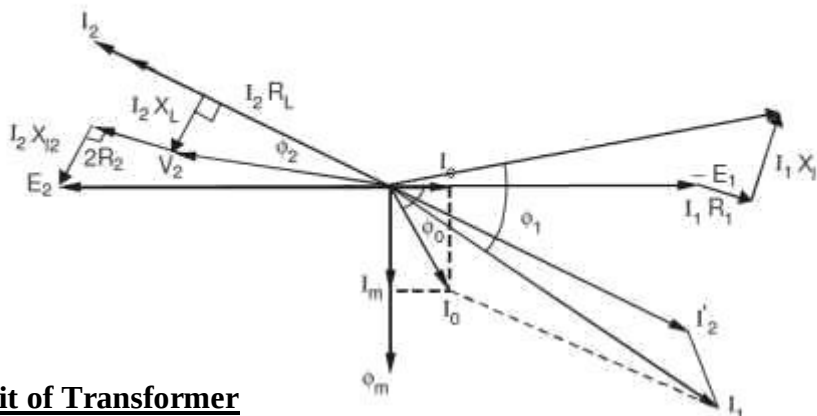
We now restrict ourselves to the more commonly occurring load i.e. inductive along with resistance,

which has a lagging power factor. For drawing this diagram, we must remember that

$$\bar{V}_2 = \bar{E}_2 - \bar{I}_2 (R_2 + j X_{L2})$$

and

$$\bar{V}_1 = -\bar{E}_1 + \bar{I}_1 (R_1 + j X_{L1})$$



Equivalent Circuit of Transformer

No load equivalent circuit

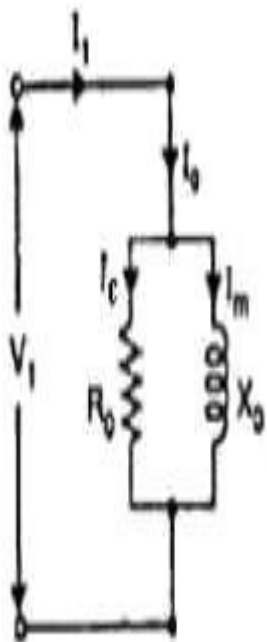


Fig:11

$$I_m = I_0 \sin \phi_0 = \text{magnetizing component}$$

$$I_c = I_0 \cos \phi_0 = \text{Active component}$$

$$R_0 = \frac{V_1}{I_c}, \quad X_0 = \frac{V_1}{I_m}$$

i) I_m produces the flux and is assumed to flow through reactance X_0 called no load reactance while I_c is active component representing core losses hence is assumed to flow through the resistance R_0

ii) Equivalent resistance is shown in fig.2.12.

iii) When the load is connected to the transformer then secondary current I_2 flows causes voltage drop across R_2 and X_2 . Due to I_2 , primary draws an additional current.

$$I_2' = \frac{I_2}{K}$$

I_1 is the phasor addition of I_0 and I_2' . This I_1 causes the voltage drop across primary resistance R_1 and reactance X_1 .

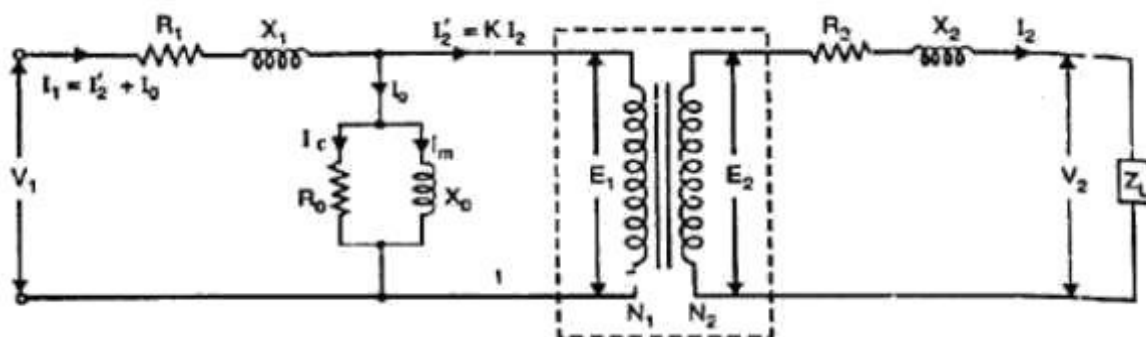


Fig: 2.12

To simplify the circuit the winding is not taken in equivalent circuit while transfer to one side.

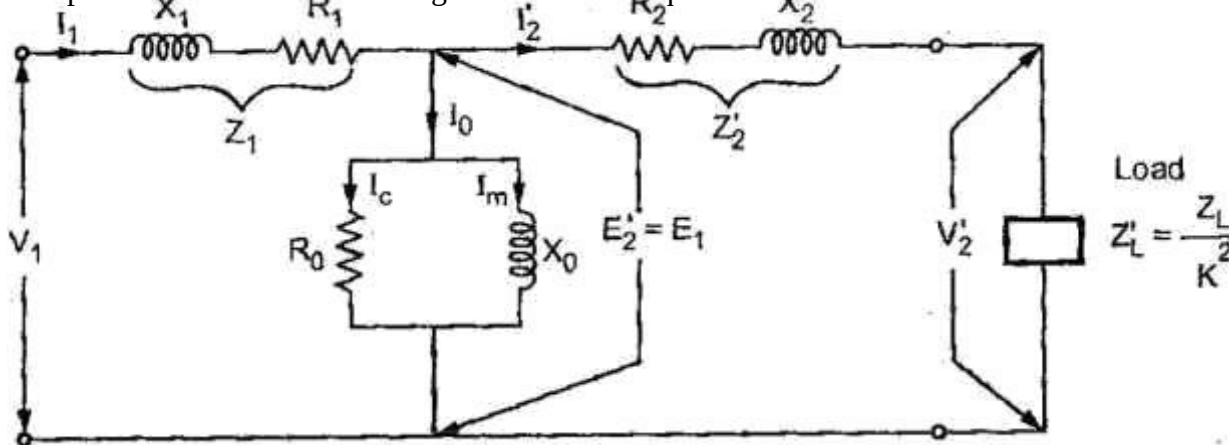


Fig: 2.13

Exact equivalent circuit referred to primary

Transferring secondary parameter to primary -

$$R_2' = \frac{R_2}{K^2}, X_2' = \frac{X_2}{K^2}, Z_2' = \frac{Z_2}{K^2}, E_2' = \frac{E_2}{K}, I_2' = KI_2, K = \frac{N_2}{N_1}$$

High voltage winding	low current	high impedance
Low voltage winding	high current	low impedance

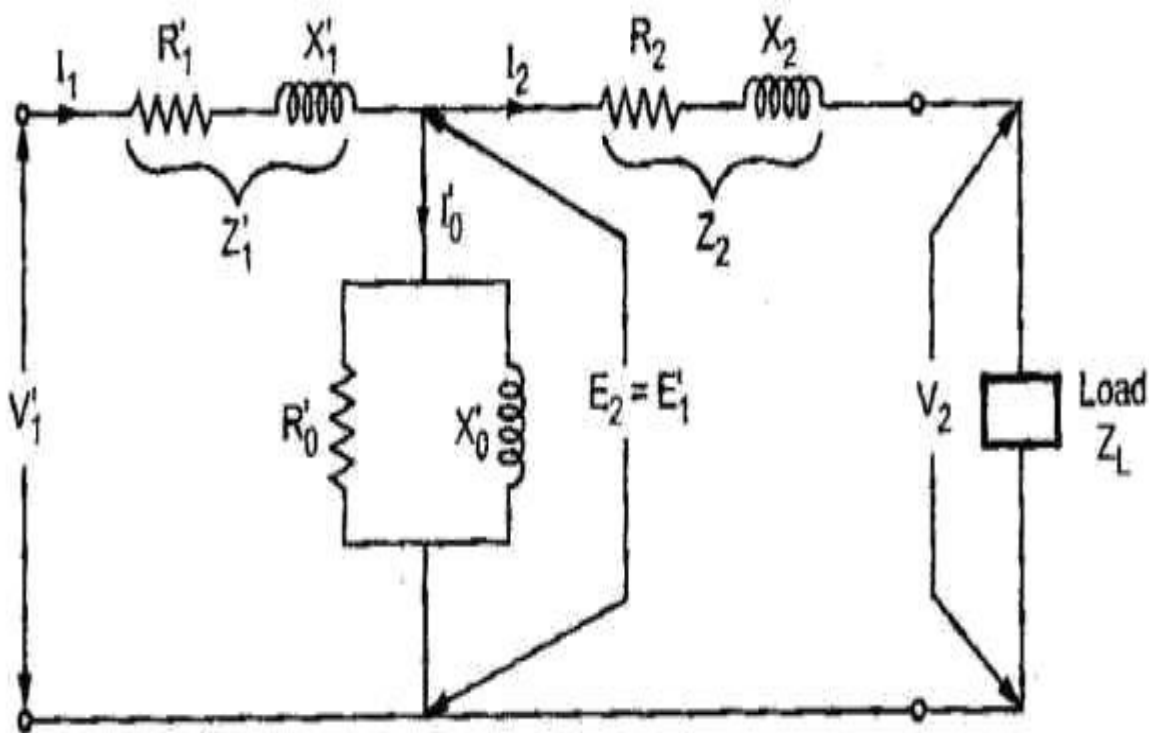


Fig: 2.14

Exact equivalent circuit referred to secondary

$$R_1' = R_1 K^2, X_1' = K^2 X_1, E_1' = K E_1$$

$$Z_1' = K^2 Z_1, I_1' = \frac{I_1}{K}, I_0 = \frac{I_0}{K}$$

Now as long as no load branch i.e. exciting branch is in between Z_1 and Z_2' , the impedances cannot be combined. So further simplification of the circuit can be done. Such circuit is called approximate equivalent circuit.

Approximate Equivalent Circuit

i) To get approximate equivalent circuit, shift the no load branch containing R_0 and X_0 to the left of R_1 and X_1 .

ii) By doing this we are creating an error that the drop across R_1 and X_1 to I_0 is neglected due to this circuit because simpler.

iii) This equivalent circuit is called approximate equivalent circuit Fig: 2.15 & Fig: 2.16.

In this circuit new R_1 and R_2' can be combined to get equivalent circuit referred to primary R_{1e} , similarly

X_1 and X_2' can be combined to get X_{1e} .

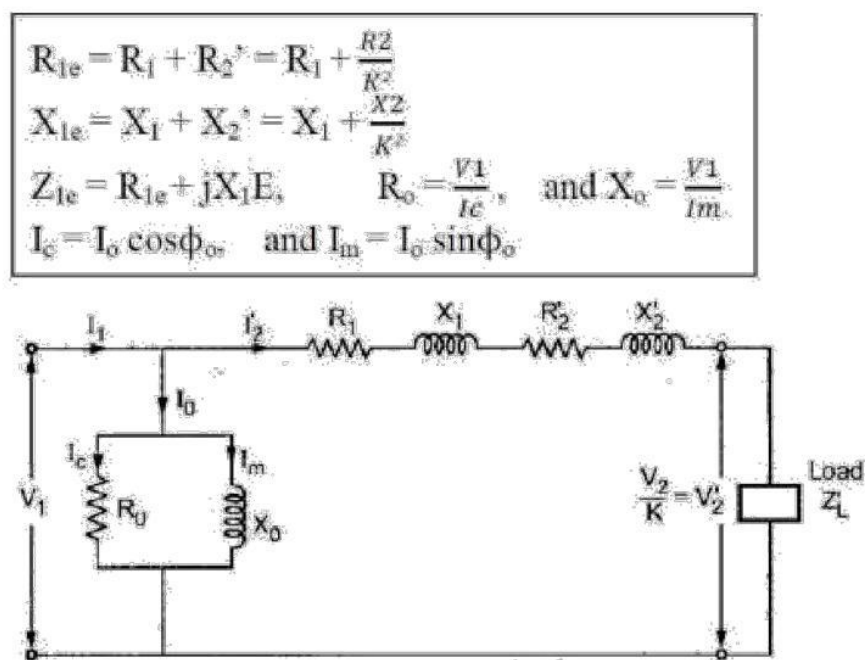
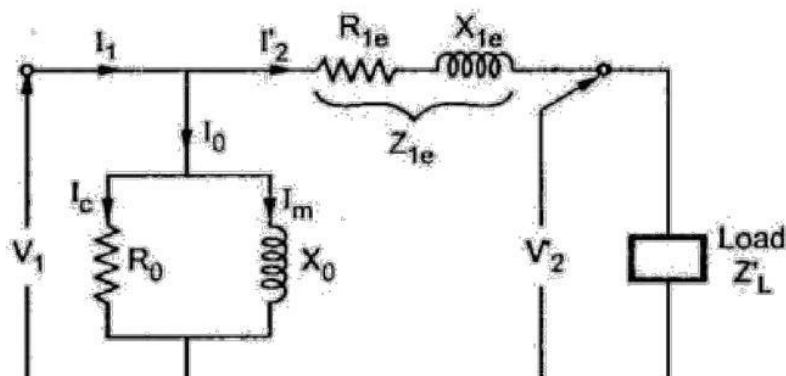


Fig: 2.15 Approximate equivalent circuit referred to primary



Approximate Voltage Drop in a Transformer

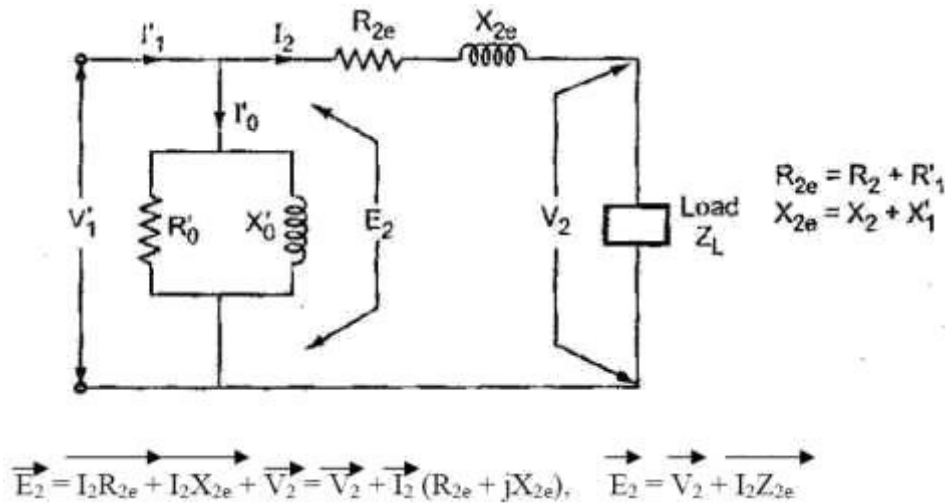


Fig. 2.17

Primary parameter is referred to secondary there are no voltage drop in primary. When there is no load,

$I_2 = 0$ and we get no load terminal voltage drop in

$$V_{20} = E_2 = \text{no load terminal voltage}$$

$$V_2 = \text{terminal voltage on load}$$

For Lagging P.F.

- i) The current I_2 lags V_2 by angle ϕ_2
- ii) Take V_2 as reference
- iii) $I_2 R_{2e}$ is in phase with I_2 while $I_2 X_{2e}$ leads I_2 by 90°
- iv) Draw the circle with O as centre and OC as radius cutting extended OA at M.
as $OA = V_2$ and now $OM = E_2$.

v) The total voltage drop is $AM = I_2 Z_{2e}$.

vi) The angle α is practically very small and in practice M&N are very close to each other. Due to this the approximate voltage drop is equal to AN instead of AM

AN – approximate voltage drop

To find AN by adding

AD & DN $AD = AB \cos \phi$

$= I_2 R_{2e} \cos \phi$ $DN = BL$

$\sin \phi = I_2 X_{2e} \sin \phi$

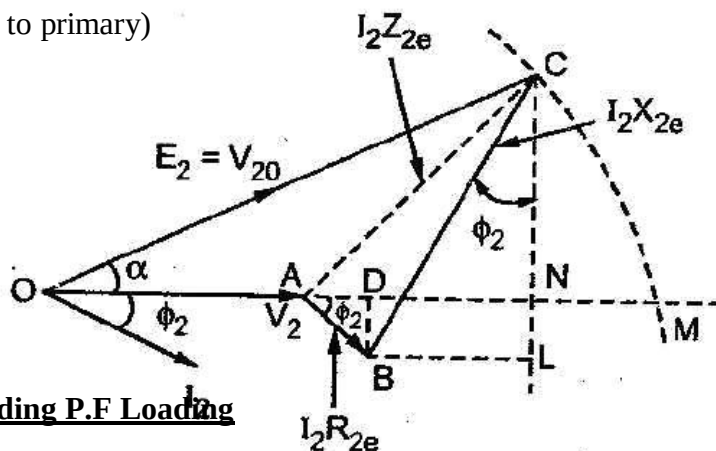
$AN = AD + DN = I_2 R_{2e} \cos \phi + I_2 X_{2e} \sin \phi$

Assuming: $\phi_2 = \phi_1 = \phi$

Approximate voltage drop = $I_2 R_{2e} \cos \phi + I_2 X_{2e} \sin \phi$ (referred to secondary)

Similarly: Approximate voltage drop = $I_1 R_{1e} \cos \phi + I_1 X_{1e} \sin \phi$

(referred to primary)



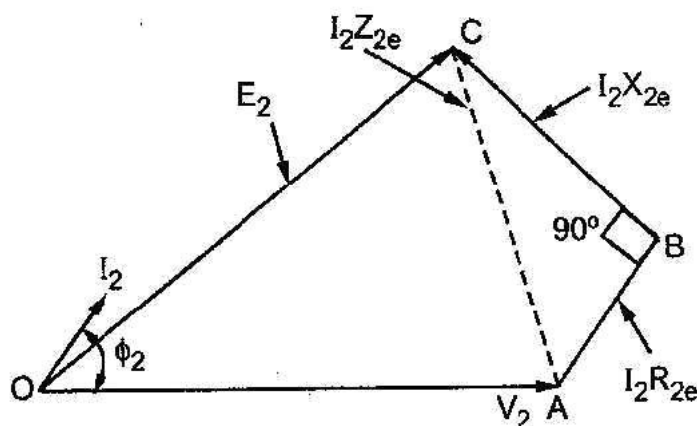
For Leading P.F Loading

I_2 leads V_2 by angle ϕ_2

Approximate voltage drop = $I_2 R_{2e} \cos \phi - I_2 X_{2e} \sin \phi$ (referred to secondary)

Similarly: Approximate voltage drop = $I_1 R_{1e} \cos \phi - I_1 X_{1e} \sin \phi$

(referred to primary)



For Unity P.F. Loading

Approximate voltage drop = $I_2 R_{2e}$ (referred to secondary)
 Similarly: Approximate voltage drop = $I_1 R_{1e}$ (referred to primary)

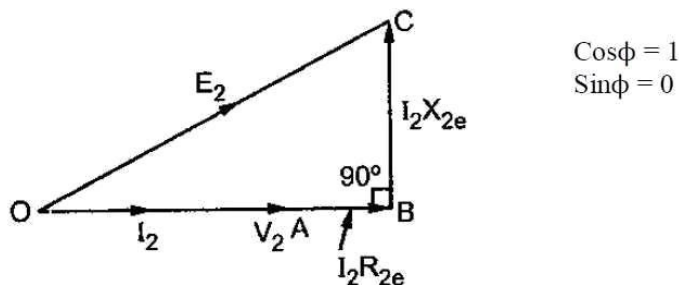


Fig: 2.20

Approximate voltage drop = $E_2 - V_2$

$$= I_2 R_{2e} \cos\phi \pm I_2 X_{2e} \sin\phi \text{ (referred to secondary)}$$

$$= I_1 R_{1e} \cos\phi \pm I_1 X_{1e} \sin\phi \text{ (referred to primary)}$$

Losses in a Transformer

The power losses in a transformer are of two types, namely;

1. Core or Iron losses
2. Copper losses

These losses appear in the form of heat and produce (i) an increase in Temperature and (ii) a drop in efficiency.

Core or Iron losses (P_i)

These consist of hysteresis and eddy current losses and occur in the transformer core due to the alternating flux. These can be determined by open-circuit test.

$$\text{Hysteresis loss} = k_h f B_m^{1.6} \text{ watts /m}^3$$

k_h – hysteresis constant depend on material

f - Frequency

B_m – maximum flux density

$$\text{Eddy current loss} = k_e f^2 B_m^2 t^2 \text{ watts /m}^3$$

K_e – eddy current constant

t - Thickness of the core

Both hysteresis and eddy current losses depend upon

(i) Maximum flux density B_m in the core

(ii) Supply frequency f . Since transformers are connected to constant-frequency, constant voltage supply, both f and B_m are constant. Hence, core or iron losses are practically the same at all loads.

Iron or Core losses, P_i = Hysteresis loss + Eddy current loss = Constant losses (P_i)

The hysteresis loss can be minimized by using steel of high silicon content. Whereas eddy current loss can be reduced by using core of thin laminations.

Copper losses (P_{cu})

These losses occur in both the primary and secondary windings due to their ohmic resistance.

These can be determined by short-circuit test. The copper loss depends on the magnitude of the current flowing through the windings.

$$\text{Total copper loss} = I_1^2 R_1 + I_2^2 R_2 = I_1^2 (R_1 + R_2') = I_2^2 (R_2 + R_1')$$

$\text{Total loss} = \text{iron loss} + \text{copper loss} = P_i + P_{cu}$
--

Efficiency of a Transformer

Like any other electrical machine, the efficiency of a transformer is defined as the ratio of output power (in watts or kW) to input power (watts or kW) i.e.

$$\text{Power output} = \text{power input} - \text{Total losses}$$

$$\text{Power input} = \text{power output} + \text{Total losses}$$

$$= \text{power output} + P_i + P_{cu}$$

$$\text{Efficiency} = \frac{\text{power output}}{\text{power input}}$$

$$\text{Efficiency} = \frac{\text{power output}}{\text{power input} + P_i + P_{cu}}$$

Power output = $V_2 I_2 \cos \phi$, $\cos \phi$ = load power factor

Transformer supplies full load of current I_2 and with terminal voltage V_2

P_{cu} = copper losses on full load = $I_2^2 R_{2e}$

This is full load efficiency and I_2 = full load current.

We can now find the full-load efficiency of the transformer at any p.f. without actually loading the transformer.

$$\text{Full load Efficiency} = \frac{(\text{Full load VA rating}) \times \cos\phi}{(\text{Full load VA rating}) \times \cos\phi + P_i + I_2^2 R_{2e}}$$

Also for any load equal to n x full-load,

$$\text{Corresponding total losses} = P_i + n^2 P_{Cu}$$

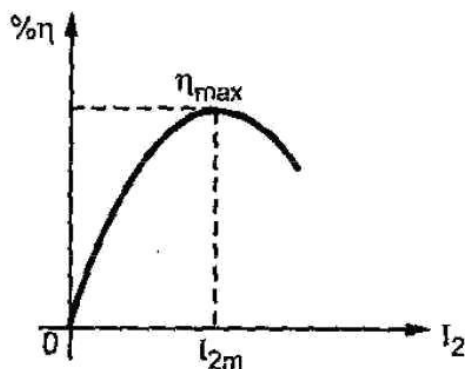
$$n = \text{fractional by which load is less than full load} = \frac{\text{actual load}}{\text{full load}}$$

$$n = \frac{\text{half load}}{\text{fullload}} = \frac{(\frac{1}{2})}{1} = 0.5$$

$$\text{Corresponding (n) \% Efficiency} = \frac{n(\text{VA rating}) \times \cos\phi}{n(\text{VA rating}) \times \cos\phi + P_i + n^2 P_{Cu}} \times 100$$

Condition for Maximum Efficiency

Voltage and frequency supply to the transformer is constant the efficiency varies with the load. As load increases, the efficiency increases. At a certain load current, it loaded further the efficiency start decreases as shown in fig. 2.21.



The load current at which the efficiency attains maximum value is denoted as I_{2m} and maximum efficiency is denoted as η_{max} , now we find -

- condition for maximum efficiency
- load current at which η_{max} occurs
- KVA supplied at maximum efficiency Considering primary side,

$$\text{Load output} = V_1 I_1 \cos \phi_1$$

$$\text{Copper loss} = I_1^2 R_{1e} \quad \text{or} \quad I_2^2 R_{2e}$$

$$\text{Iron loss} = \text{hysteresis} + \text{eddy current loss} = P_i$$

$$\begin{aligned} \text{Efficiency} &= \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1} = \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{1e} + P_i}{V_1 I_1 \cos \phi_1} \\ &= 1 - \frac{I_1 R_{1e}}{V_1 I_2 \cos \phi_1} = \frac{P_i}{V_1 I_1 \cos \phi_1} \end{aligned}$$

Differentiating both sides with respect to I_2 , we get

$$\frac{d\eta}{dI_2} = 0 - \frac{R_{1e}}{V_1 \cos \phi_1} = \frac{P_i}{V_1 I_1^2 \cos \phi_1}$$

For η to be maximum, $\frac{d\eta}{dI_2} = 0$. Hence, the above equation becomes

$$\frac{R_{1e}}{V_1 \cos \phi_1} = \frac{P_i}{V_1 I_1^2 \cos \phi_1} \quad \text{or} \quad P_i = I_1^2 R_{1e}$$

$$P_{cu} \text{ loss} = P_i \text{ iron loss}$$

The output current which will make P_{cu} loss equal to the iron loss. By proper design, it is possible to make the maximum efficiency occur at any desired load.

Load current I_{2m} at maximum efficiency

KVA Supplied at Maximum Efficiency

For constant V_2 the KVA supplied is the function of load current.

$$\text{For } \eta_{\max} \quad I_2^2 R_{2e} = P_i \text{ but } I_2 = I_{2m}$$

$$I_{2m}^2 R_{2e} = P_i \quad I_{2m} = \sqrt{\frac{P_i}{R_{2e}}}$$

This is the load current at $\eta_{\max}$.
(I_2) F.L. = full load current

$$\frac{I_{2m}}{(I_2) \text{ F.L.}} = \frac{1}{(I_2) \text{ F.L.}} \sqrt{\frac{P_i}{R_{2e}}}$$

$$\frac{I_{2m}}{(I_2) \text{ F.L.}} = \sqrt{\frac{P_i}{[(I_2) \text{ F.L.}]^2 R_{2e}}} = \sqrt{\frac{P_i}{[P_{cu}] \text{ F.L.}}}$$

$$I_{2m} = (I_2) \text{ F.L.} \sqrt{\frac{P_i}{[P_{cu}] \text{ F.L.}}}$$

This is the load current at $\eta_{\max}$ in terms of full load current

KVA Supplied at Maximum Efficiency

For constant V_2 the KVA supplied is the function of load current.

$$\begin{aligned} \text{KVA at } \eta_{\max} &= I_{2m} V_2 = V_2 (I_2) \text{ F.L.} \times \sqrt{\frac{P_i}{[P_{cu}] \text{ F.L.}}} \\ \text{KVA at } \eta_{\max} &= (\text{KVA rating}) \times \sqrt{\frac{P_i}{[P_{cu}] \text{ F.L.}}} \end{aligned}$$

Substituting condition for $\eta_{\max}$ in the expression of efficiency, we can write expression for $\eta_{\max}$ as ,

as $P_{cu} = P_i$

$$\% \eta_{\max} = \frac{V_2 I_{2m} \cos \phi}{V_2 I_{2m} \cos \phi + 2P_i} \times 100$$

All Day Efficiency (Energy Efficiency)

In electrical power system, we are interested to find out the all-day efficiency of any transformer because the load at transformer is varying in the different time duration of the day. So all day efficiency is defined as the ratio of total energy output of transformer to the total energy input in 24 hours.

$$\text{All day efficiency} = \frac{\text{kWh output during a day}}{\text{kWh input during the day}}$$

UNIT-II

TESTING OF TRANSFORMERS

Testing of Transformer

The testing of transformer means to determine efficiency and regulation of a transformer at any load and at any power factor condition.

There are two methods

- i) Direct loading test
- ii) Indirect loading test

a. Open circuit test

b. Short circuit test

i) Load test on transformer

This method is also called as direct loading test on transformer because the load is directly connected to the transformer. We required various meters to measure the input and output reading while change the load from zero to full load. Fig. 2.22 shows the connection of transformer for direct load test. The primary is connected through the variac to change the input voltage as we required. Connect the meters as shown in the figure below.

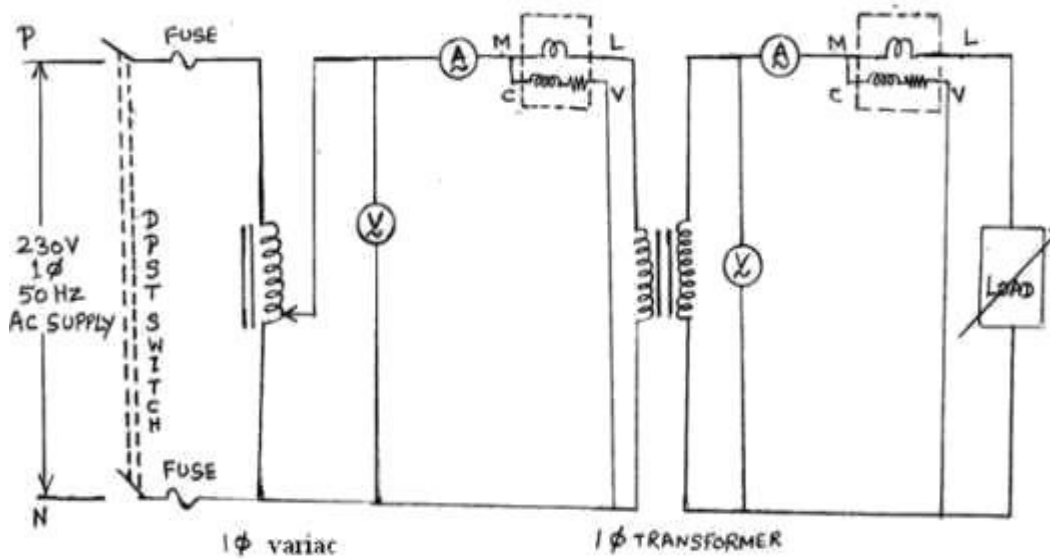


Fig: 2.22

The load is varied from no load to full load in desired steps. All the time, keep primary voltage V_1 constant at its rated value with help of variac and tabulated the reading. The first reading is to be noted on no load for which $I_2 = 0$ A and $W_2 = 0$ W.

Calculation

From the observed reading

W_1 = input power to the transformer

W_2 = output power delivered to the load

$$\% \eta = \frac{W_2}{W_1} \times 100$$

The first reading is no load so $V_2 = E_2$

The regulation can be obtained as

$$\% R = \frac{E_2 - V_2}{E_2} \times 100$$

The graph of % η and % R on each load against load current I_L is plotted as shown in fig. 2.23.

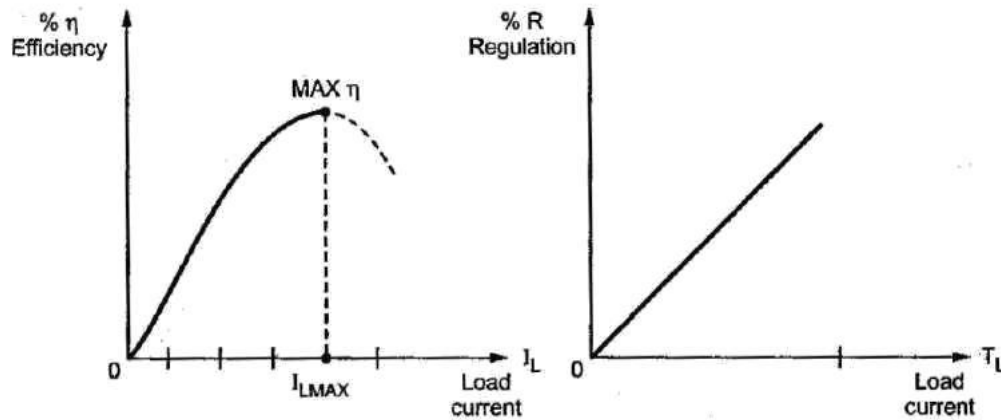


Fig: 2.23

Advantages:

- 1) This test enables us to determine the efficiency of the transformer accurately at any load.
- 2) The results are accurate as load is directly used.

Disadvantages:

- 1) There are large power losses during the test.
- 2) Load not avail in lab while test conduct for large transformer.

ii) a. Open-Circuit or No-Load Test

This test is conducted to determine the iron losses (or core losses) and parameters R_0 and X_0 of the transformer. In this test, the rated voltage is applied to the primary (usually low-voltage

winding) while the secondary is left open circuited. The applied primary voltage V_1 is measured by the voltmeter, the no load current I_0 by ammeter and no-load input power W_0 by wattmeter as shown in Fig.2.24.a. As the normal rated voltage is applied to the primary, therefore, normal iron losses will occur in the transformer core. Hence wattmeter will record the iron losses and small copper loss in the primary. Since no-load current I_0 is very small (usually 2-10 % of rated current). Cu losses in the primary under no-load condition are negligible as compared with iron losses. Hence, wattmeter reading practically gives the iron losses in the transformer. It is reminded that iron losses are the same at all loads.

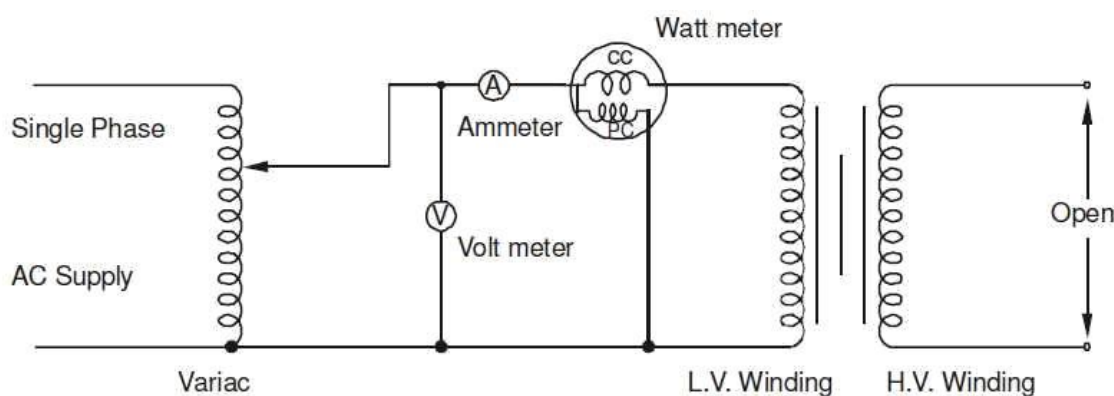


Fig: 2.24.a

Iron losses, P_i = Wattmeter reading = W_0

No load current = Ammeter reading = I_0

Applied voltage = Voltmeter reading = V_1

Input power, $W_0 = V_1 I_0 \cos \phi_0$

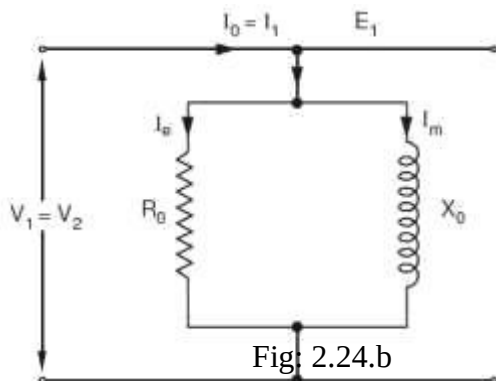
No - load p.f., $\cos \phi = \frac{W_0}{V_0 I_0}$ = no load power factor

$I_m = I_0 \sin \phi_0$ = magnetizing component

$I_c = I_0 \cos \phi_0$ = Active component

$$R_0 = \frac{V_0}{I_c} \Omega, \quad X_0 = \frac{V_0}{I_m} \Omega$$

Under no load conditions the PF is very low (near to 0) in lagging region. By using the above data we can draw the equivalent parameter shown in Figure 2.24.b.



Thus open-circuit test enables us to determine iron losses and parameters R_0 and X_0 of the transformer

ii) b. Short-Circuit or Impedance Test

This test is conducted to determine R_{1e} (or R_{2e}), X_{1e} (or X_{2e}) and full-load copper losses of the transformer. In this test, the secondary (usually low-voltage winding) is short-circuited by a thick conductor and variable low voltage is applied to the primary as shown in Fig.2.25. The low input voltage is gradually raised till at voltage V_{SC} , full-load current I_1 flows in the primary. Then I_2 in the secondary also has full-load value since $I_1/I_2 = N_2/N_1$. Under such conditions, the copper loss in the windings is the same as that on full load. There is no output from the transformer under short-circuit conditions. Therefore, input power is all loss and this loss is almost entirely copper loss. It is because iron loss in the core is negligibly small since the voltage V_{SC} is very small. Hence, the wattmeter will practically register the full load copper losses in the transformer windings.

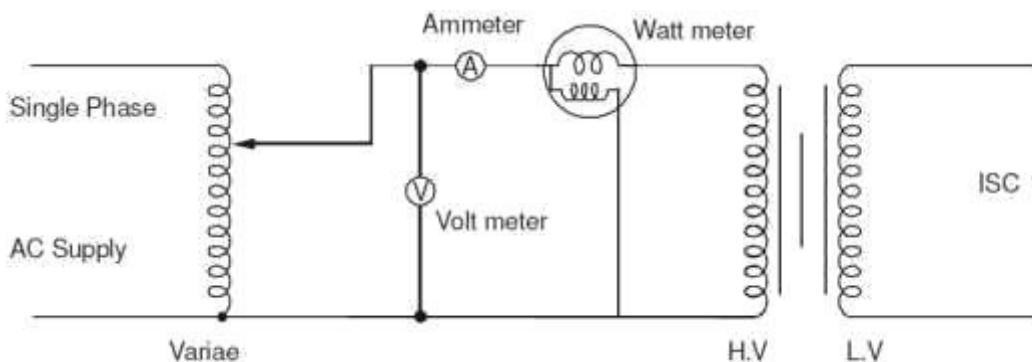


Fig: 2.25.a

Full load Cu loss, $P_C = \text{Wattmeter reading} = W_{sc}$

Applied voltage = Voltmeter reading = V_{sc}

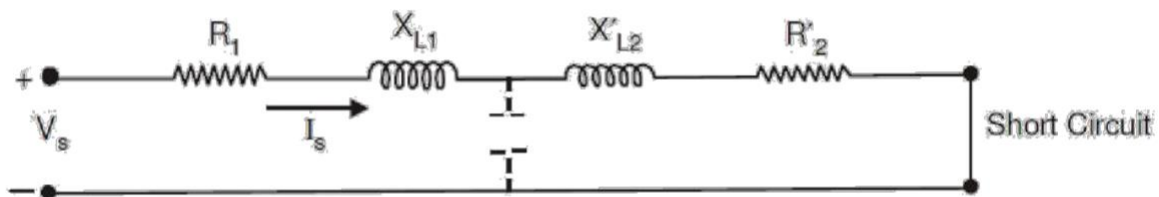
F.L. primary current = Ammeter reading = I_1

$$P_{Cu} = I_1^2 R_1 + I_1^2 R_2' = I_1^2 R_{1e}, \quad R_{1e} = \frac{P_{Cu}}{I_1^2}$$

Where R_{1e} is the total resistance of transformer referred to primary.

Total impedance referred to primary, $Z_{1e} = \sqrt{Z_{1e}^2 - R_{1e}^2}$

short-circuit P.F., $\cos \Phi = \frac{P_{Cu}}{V_{sc} I_1}$ Thus short-circuit test gives full-load Cu loss, R_{1e} and X_{1e} .



$$\text{equivalent resistance } R_{eq} = \frac{W_s}{I_s^2} = R_1 + R_2'$$

$$\text{and equivalent impedance } Z_{eq} = \frac{V_s}{I_s}$$

So we calculate equivalent reactance

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = X_{L1} + X_{L2}'$$

These R_{eq} and X_{eq} are equivalent resistance and reactance of both windings referred in HV side. These are known as equivalent circuit resistance and reactance.

Voltage Regulation of Transformer

Under no load conditions, the voltage at the secondary terminals is E_2 and

$$E_2 \approx V_1 \cdot \frac{N_2}{N_1}$$

(This approximation neglects the drop R_1 and X_{L1} due to small no load current). As load is applied

to the transformer, the load current or the secondary current increases. Correspondingly, the primary current I_1 also increases. Due to these currents, there is a voltage drop in the primary and secondary leakage reactances, and as a consequence the voltage across the output terminals or the load terminals changes. In quantitative terms this change in terminal voltage is called Voltage Regulation.

Voltage regulation of a transformer is defined as the drop in the magnitude of load voltage (or secondary terminal voltage) when load current changes from zero to full load value. This is expressed as a fraction of secondary rated voltage.

$$\text{Regulation} = \frac{\text{Secondary terminal voltage at no load} - \text{Secondary terminal voltage at any load}}{\text{Secondary rated voltage}}$$

The secondary rated voltage of a transformer is equal to the secondary terminal voltage at no load (i.e. E_2), this is as per IS.

Voltage regulation is generally expressed as a percentage.

$$\text{Percent voltage regulation (\% VR)} = \frac{E_2 - V_2}{E_2} \times 100.$$

Note that E_2 , V_2 are magnitudes, and not phasor or complex quantities. Also note that voltage regulation depends not only on load current, but also on its power factor. Using approximate equivalent circuit referred to primary or secondary, we can obtain the voltage regulation. From approximate equivalent circuit referred to the secondary side and phasor diagram for the circuit.

$$E_2 = V_2 + I_2 r_{eq} \cos \phi_2 \pm I_2 x_{eq} \sin \phi_2$$

where $r_{eq} = r_2 + r_1'$ (referred to secondary) $x_e = x_2 + x_1'$ (+ sign applies lagging power factor load and – sign applies to leading pf load).

$$\text{So } \frac{E_2 - V_2}{E_2} = \frac{I_2 r_{eq} \cos \phi_2 \pm I_2 x_{eq} \sin \phi_2}{E_2}$$

$$\frac{E_2 - V_2}{E_2} = \frac{I_2 r_{eq}}{E_2} \cos \phi_2 \pm \frac{I_2 x_{eq}}{E_2} \sin \phi_2$$

$$\% \text{ Voltage regulation} = (\% \text{ resistive drop}) \cos \phi_2 \pm (\% \text{ reactive drop}) \sin \phi_2.$$

Ideally voltage regulation should be zero.

UNIT-III
Auto Transformer & Parallel Operation

Auto-transformers

The transformers we have considered so far are two-winding transformers in which the electrical circuit connected to the primary is electrically isolated from that connected to the secondary. An auto-transformer does not provide such isolation, but has economy of cost combined with increased efficiency. Fig.2.26 illustrates the auto-transformer which consists of a coil of N_A turns between terminals 1 and 2, with a third terminal 3 provided after N_B turns. If we neglect coil resistances and leakage fluxes, the flux linkages of the coil between 1 and 2 equals $N_A \phi_m$ while the portion of coil between 3 and 2 has a flux linkage $N_B \phi_m$. If the induced voltages are designated as E_A and E_B , just as in a two winding transformer,

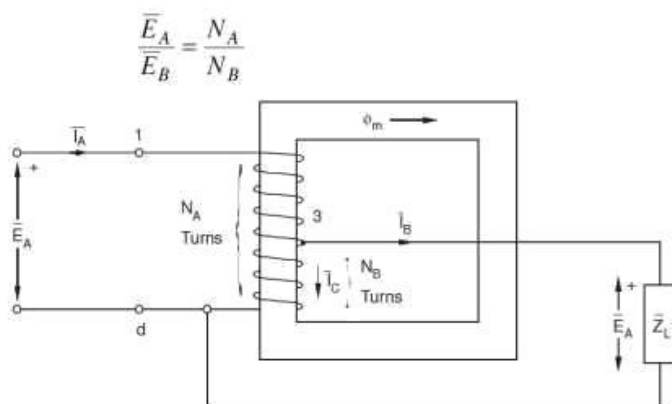


Fig: 2.2

Neglecting the magnetizing ampere-turns needed by the core for producing flux, as in an ideal transformer, the current I_A flows through only $(N_A - N_B)$ turns. If the load current is I_B , as shown by

Kirchhoff's current law, the current I_C flowing from terminal 3 to terminal 2 is $(I_A - I_B)$. This current flows through N_B turns. So, the requirement of a net value of zero ampere-turns across the core demands that

$$(N_A - N_B) \bar{I}_A + (\bar{I}_A - \bar{I}_B) N_B = 0$$

$$\text{or} \quad N_A \bar{I}_A - N_B \bar{I}_B = 0$$

Hence, just as in a two-winding transformer,

$$\frac{\bar{I}_A}{\bar{I}_B} = \frac{N_B}{N_A}$$

Consequently, as far as voltage, current converting properties are concerned, the autotransformer of Figure: 26 behaves just like a two-winding transformer. However, in the autotransformer we don't need two separate coils, each designed to carry full load values of current.

Parallel Operation of Transformers

It is economical to install numbers of smaller rated transformers in parallel than installing bigger rated electrical power transformers. This has mainly the following advantages,

To maximize electrical power system efficiency: Generally electrical power transformer gives the maximum efficiency at full load. If we run numbers of transformers in parallel, we can switch on only those transformers which will give the total demand by running nearer to its full load rating for that time. When load increases, we can switch none by one other transformer connected in parallel to fulfill the total demand. In this way we can run the system with maximum efficiency.

To maximize electrical power system availability: If numbers of transformers run in parallel, we can shut down any one of them for maintenance purpose. Other parallel transformers in system will serve the load without total interruption of power.

To maximize power system reliability: if any one of the transformers run in parallel, is tripped due to fault of other parallel transformers is the system will share the load, hence power supply may not be interrupted if the shared loads do not make other transformers over loaded.

To maximize electrical power system flexibility: There is always a chance of increasing or decreasing future demand of power system. If it is predicted that power demand will be increased in future, there must be a provision of connecting transformers in system in parallel to fulfil the extra demand because, it is not economical from business point of view to install a bigger rated single transformer by forecasting the increased future demand as it is unnecessary investment of money. Again if future demand is decreased, transformers running in parallel can be removed from system to balance the capital investment and its return.

Conditions for Parallel Operation of Transformers

When two or more transformers run in parallel, they must satisfy the following conditions for satisfactory performance. These are the conditions for parallel operation of transformers.

- ☐ *Same voltage ratio of transformer.*
- ☐ *Same percentage impedance.*
- ☐ *Same polarity.*
- ☐ *Same phase sequence.*

- ☐ *Same Voltage Ratio*

Same voltage ratio of transformer

If two transformers of different voltage ratio are connected in parallel with same primary supply voltage, there will be a difference in secondary voltages. Now say the secondary of these transformers are connected to same bus, there will be a circulating current between secondary's and therefore between primaries also. As the internal impedance of transformer is small, a small voltage difference may cause sufficiently high circulating current causing unnecessary extra I²R loss.

Same Percentage Impedance

The current shared by two transformers running in parallel should be proportional to their MVA ratings. Again, current carried by these transformers are inversely proportional to their internal impedance. From these two statements it can be said that, impedance of transformers running in parallel are inversely proportional to their MVA ratings. In other words, percentage impedance or per unit values of impedance should be identical for all the transformers that run in parallel.

Same Polarity

Polarity of all transformers that run in parallel, should be the same otherwise huge circulating current that flows in the transformer but no load will be fed from these transformers. Polarity of transformer means the instantaneous direction of induced emf in secondary. If the instantaneous

directions of induced secondary emf in two transformers are opposite to each other when same input power is fed to both of the transformers, the transformers are said to be in opposite polarity. If the instantaneous directions of induced secondary e.m.f in two transformers are same when same input power is fed to the both of the transformers, the transformers are said to be in same polarity.

Same Phase Sequence

The phase sequence or the order in which the phases reach their maximum positive voltage, must be identical for two parallel transformers. Otherwise, during the cycle, each pair of phases will be short circuited.

The above said conditions must be strictly followed for parallel operation of transformers but totally identical percentage impedance of two different transformers is difficult to achieve practically, that is why the transformers run in parallel may not have exactly same percentage impedance but the values would be as nearer as possible.

Why Transformer Rating in kVA?

An important factor in the design and operation of electrical machines is the relation between the life of the insulation and operating temperature of the machine. Therefore, temperature rise resulting from the losses is a determining factor in the rating of a machine. We know that copper loss in a transformer depends on current and iron loss depends on voltage. Therefore, the total loss in a transformer depends on the volt-ampere product only and not on the phase angle between voltage and current i.e., it is independent of load power factor. For this reason, the rating of a transformer is in kVA and not kW.

UNIT-IV
Poly Phase Induction Machines

Introduction

The induction machine was invented by NIKOLA TESLA in 1888. Right from its inception its ease of manufacture and its robustness have made it a very strong candidate for electromechanical energy conversion. It is available from fractional horsepower ratings to megawatt levels. It finds very wide usage in all various application areas. The induction machine is an AC electromechanical energy conversion device. The machine interfaces with the external world through two connections (ports) one mechanical and one electrical. The mechanical port is in the form of a rotating shaft and the electrical port is in the form of terminals where AC supply is connected. There are machines available to operate from three phase or single phase electrical input. In this module we will be discussing the three phase induction machine. Single phase machines are restricted to small power levels.

The Rotating Magnetic Field

The principle of operation of the induction machine is based on the generation of a rotating magnetic field. Let us understand this idea better.

Click on the following steps in sequence to get a graphical picture. It is suggested that the reader read the text before clicking the link.

- Consider a cosine wave from 0 to 360°. This sine wave is plotted with unit amplitude.
- Now allow the amplitude of the sine wave to vary with respect to time in a sinusoidal fashion with a frequency of 50Hz. Let the maximum value of the amplitude is, say, 10 units. This waveform is a pulsating sine wave.

$$i_{apk} = I_m \cos 2\pi \cdot 50 \cdot t \quad (1)$$

- Now consider a second sine wave, which is displaced by 120° from the first (lagging). . .
- and allow its amplitude to vary in a similar manner, but with a 120° time lag.

$$i_{bpk} = I_m \cos(2\pi \cdot 50 \cdot t - 120^\circ) \quad (2)$$

- Similarly consider a third sine wave, which is at 240° lag. . .
- and allow its amplitude to change as well with a 240° time lag. Now we have three pulsating sine waves.

$$i_{cpk} = I_m \cos(2\pi \cdot 50 \cdot t - 240^\circ) \quad (3)$$

Let us see what happens if we sum up the values of these three sine waves at every angle. The result really speaks about Tesla's genius. What we get is a constant amplitude travelling sine wave!

In a three phase induction machine, there are three sets of windings — phase A winding, phase B and phase C windings. These are excited by a balanced three-phase voltage supply. This would result in a balanced three phase current. Equations 1 — 3 represent the currents that flow in the three phase windings. Note that they have a 120° time lag between them.

Further, in an induction machine, the windings are not all located in the same place. They are distributed in the machine 120° away from each other (more about this in the section on alternators). The correct terminology would be to say that the windings have their axes separated in space by 120° . This is the reason for using the phase A, B and C since waves separated in space as well by 120° .

When currents flow through the coils, they generate mmfs. Since mmf is proportional to current, these waveforms also represent the mmf generated by the coils and the total mmf. Further, due to magnetic material in the machine (iron), these mmfs generate magnetic flux, which is proportional to the mmf (we may assume that iron is infinitely permeable and non-linear effects such as hysteresis are neglected). Thus the waveforms seen above would also represent the flux generated within the machine. The net result as we have seen is a travelling flux wave. The x-axis would represent the space angle in the machine as one travels around the air gap. The first pulsating waveform seen earlier would then represent the a-phase flux, the second represents the b-phase flux and the third represents the c-phase.

This may be better visualized in a polar plot. The angles of the polar plot represent the space angle in the machine, i.e., angle as one travels around the stator bore of the machine. Click on the links below to see the development on a polar axes.

- This plot shows the pulsating wave at the zero degree axes. The amplitude is maximum at zero degree axes and is zero at 90° axis. Positive parts of the waveform are shown in red while negative in blue. Note that the waveform is pulsating at the $0 - 180^\circ$ axis and red and blue alternate in any given side. This corresponds to the sinewave current

changing polarity. Note that the maximum amplitude of the sinewave is reached only along the $0 - 180^\circ$ axis. At all other angles, the amplitude does not reach a maximum of this value. It however reaches a maximum value which is less than that of the peak occurring at the $0 - 180^\circ$ axis. More exactly, the maximum reached at any space angle θ would be equal to $\cos\theta$ times the peak at the $0 - 180^\circ$ axis. Further, at any space angle θ , the time variation is sinusoidal with the frequency and phase lag being that of the excitation, and amplitude being that corresponding to the space angle.

- This plot shows the pulsating waveforms of all three cosines. Note that the first is pulsating about the $0 - 180^\circ$ axis, the second about the $120^\circ - 300^\circ$ axis and the third at $240^\circ - 360^\circ$ axis.
- This plot shows the travelling wave in a circular trajectory. Note that while individual pulsating waves have maximum amplitude of 10, the resultant has amplitude of 15.

If f_1 is the amplitude of the flux waveform in each phase, the travelling wave can

then be represented as

$$\begin{aligned}
 f(t) &= f_1 \cos \omega t \cos \theta + f_1 \cos\left(\omega t - \frac{2\pi}{3}\right) \cos\left(\theta - \frac{2\pi}{3}\right) + f_1 \cos\left(\omega t - \frac{4\pi}{3}\right) \cos\left(\theta - \frac{4\pi}{3}\right) \\
 &= 2 f_1 \cos(\omega t - \theta)
 \end{aligned} \tag{4}$$

Principles of Torque Production

In the earlier section, we saw how a rotating flux is produced. Now let us consider a rotor, which is placed in this field. Let the rotor have a coil such that the coil sides are placed diametrically opposite each other. This is shown in the fig. 1. Since the flux generated by the stator rotates flux linked by this rotor coil also changes.

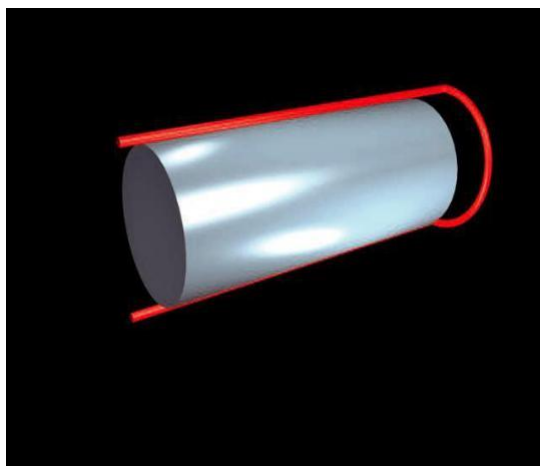


Figure 1: A Coil on the rotor

Since the flux pattern is varying sinusoidally in space, as the flux waveform rotates, the flux linkage varies sinusoidally. The rate of variation of this flux linkage will then be equal to the speed of rotation of the air gap flux produced. This sinusoidal variation of the flux linkage produces a sinusoidal induced emf in the rotor coil. If the coil is short circuited, this induced emf will cause a current flow in the coil as per Lenz's law.

Now imagine a second coil on the rotor whose axis is 120° away from the first. This is shown in fig. 2. The flux linkage in this coil will also vary sinusoidally with respect to time and therefore cause an induced voltage varying sinusoidally with time. However the flux linkages in these two coils will have a phase difference of 120° (the rotating flux wave will have to travel 120° in order to cause a similar flux linkage variation as in the first coil), and hence the time varying voltages induced in the coils will also have a 120° phase difference.

A third coil placed a further 120° away is shown in fig. 3. This will have a time varying induced emf lagging 240° in time with respect to the first.

When these three coils are shorted upon themselves currents flow in them as per Lenz's law. The mechanism by which torque is produced may now be understood as follows. The diagram in fig. 4 shows a view of the rotor seen from one end. Positive current is said to

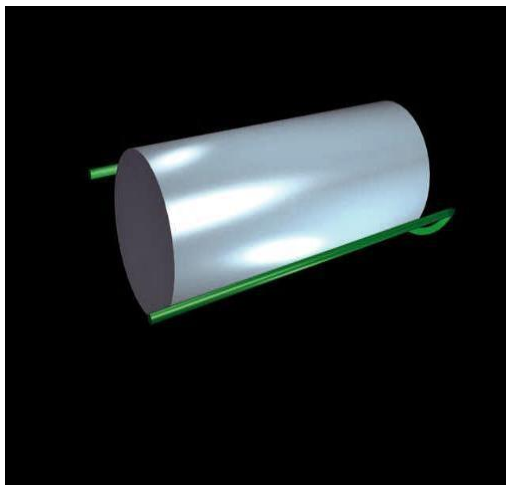


Figure 2: A coil displaced 120° from the first

flow in these coils when current flows out of the page in a, b, c conductors and into a' , b' and c' respectively.

If we look at the voltage induced in these coils as phasors, the diagram looks as shown in fig. 5. The main flux is taken as the reference phasor. Considering that the induced emf is $-d\psi/dt$ where ψ is the flux linkage, the diagram is drawn as shown.

As usual, the horizontal component of these phasors gives the instantaneous values of the induced emf in these coils.

Let these coils be purely resistive. Then these emf phasors also represent the currents flowing in these coils. If we consider the instant $t = 0$, it can be seen that

1. The field flux is along 0° axis.
2. The current in a phase coil is zero.
3. The current in c phase coil is $+2^3$ units.

These currents act to produce mmf and flux along the axes of the respective coils. Let us consider

the space around b' and c coil sides. The situation is shown in fig. 6.

The resulting flux pattern causes a tendency to move in the anticlockwise direction. This is easy to see through the so called whiplash rule. Alternatively, since the force on a current

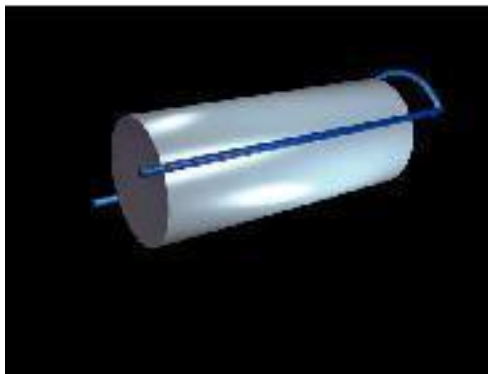


Figure 3: A coil displaced 240° from the first

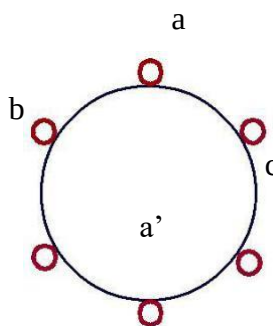


Figure 4: Coils on the rotor

carrying conductor is $F = q(v \times B)$, it can be seen that the torque produced tends to rotate the rotor counter-clockwise. The magnitude of the torque would increase with the current magnitude in the coils. This current is in turn dependent on the magnitude of the main field flux and its speed of rotation. Therefore one may say that motion of the main field tends to drag the rotor along with it.

When the rotor is free to move and begins moving, the motion reduces the relative speed between the main field and the rotor coils. Less emf would therefore be induced and the torque would come down. Depending on the torque requirement for the load, the difference in speed between the rotor and the main field settles down at some particular value.

From the foregoing, the following may be noted.

1. The torque produced depends on a non-zero relative speed between the field and the rotor.

2. It is therefore not possible for the rotor to run continuously at the same speed of the field. This is so because in such a condition, no emf would be induced in the rotor and hence no rotor current, no torque.
3. The frequency of currents induced in the rotor coils and their magnitude depends on this difference in speed.

These are important conclusions. The speed of the main field is known as the synchronous speed, n_s . If the actual speed of the rotor is n_r then the ratio is known as slip and is frequently expressed as a percentage. Typically induction machines are designed to operate at about less than 4 percent slip at full load.

$$s = \frac{n_s - n_r}{n_s} \quad (5)$$

It is instructive to see the situation if the rotor resistance is neglected and is considered to be purely inductive. The phasor diagram of voltages and the currents would then look as shown in fig. 7.

At $t = 0$, one can see that current in a phase coil is at negative maximum, while b and c phases have positive current of 0.5 units. Now if we consider the current flux profiles at coil sides a, b', c, the picture that emerges is shown in fig. 8.

Since main flux at the a coil side is close to zero, there is very little torque produced from there. There is a tendency to move due to the b' and c coil sides, but they are in opposite directions however. Hence there is no net torque on the rotor. This brings up another important conclusion — the resistance of the rotor is an important part of torque production in the induction machine. While a high resistance rotor is better suited for torque production, it would also be lossy.

Construction

In actual practice, the three coils form three windings distributed over several slots. These windings may be connected in star or delta and three terminations are brought out. These are conventional three phase windings which are discussed in greater detail in the chapters on alternators. Such windings are present in the stator as well as rotor. A photograph of



Figure 9: stator of an induction machine

the stator of an induction machine is shown in fig. 9. A close up of the windings is shown in fig. 10. the several turns that make up a coil are seen in this picture. The three terminations are connected to rings on which three brushes make a sliding contact. As the rotor rotates the brushes slip over the rings and provide means of connecting stationary external circuit elements to the rotating windings. A schematic of these arrangements is shown in fig. 13. A photograph of a wound rotor for an induction machine is shown in fig. 11. Fig. 12 shows a close up of the slip ring portion. Brushes are not shown in this picture.

Induction machines, which have these kinds of windings and terminals that are brought out, are called slip ring machines. The reader may note that in order that torque is produced current must flow in the rotor. To achieve that, the stationary brush terminals must either be shorted, or connected to a circuit allowing current flow. Sometimes a star connected resistor bank is connected so that the developed starting torque is higher. There are also other forms of power electronic circuitry that may be connected to the rotor terminals to achieve various functions. The popularity of the induction machine however, stems from another variety of rotor



Figure 10: Coils in the stator



Figure 11: A wound rotor with slip rings



Figure 12: slip rings

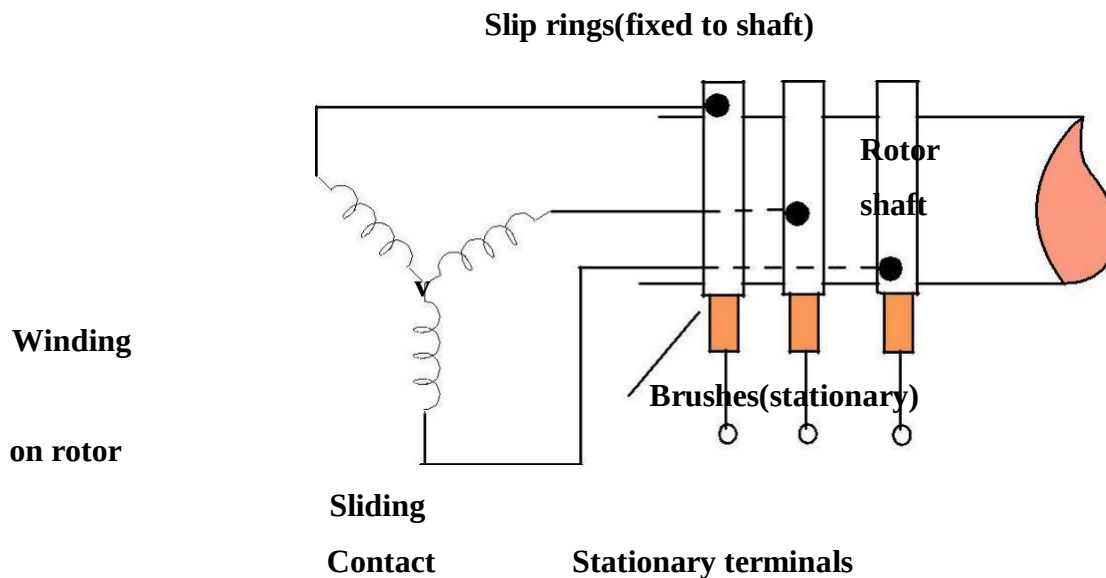


Figure 13: Slip rings and brushes in induction machines — a schematic That is used. This rotor has slots into which copper or aluminum bars are inserted. These bars are then shorted by rings that are brazed on to each of the rotor ends. Figure 14 shows a simple schematic.

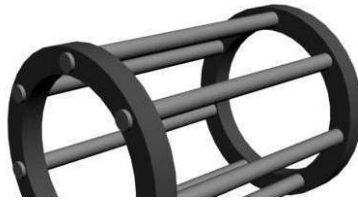


Figure 14: Squirrel cage rotor — a schematic

Such a rotor is called squirrel cage rotor. This rotor behaves like a short-circuited winding and hence the machine is able to perform electromechanical energy conversion. This type of rotor is easy to manufacture, has no sliding contacts and is very robust. It is this feature that makes induction machine suitable for use even in hazardous environments and reliable operation is achieved. The disadvantage of this type of rotor is that the motor behavior cannot be altered by connecting anything to the rotor — there are no rotor terminals.

Fig. 15 shows a photograph of a squirrel cage rotor. The rotor also has a fan attached to it. This is for cooling purposes. The bars (white lines on the surface) are embedded in the rotor iron which forms the magnetic circuit. The white lines correspond to the visible portion of the rotor bar.

Sometimes two rotor bars are used per slot to achieve some degree of variability in the starting and running performances. It is to make use of the fact that while high rotor



Figure 15: squirrel cage rotor

resistance is desirable from the point of view of starting torque, low rotor resistance is desirable

from efficiency considerations while the machine is running. Such rotors are called double cage rotors or deep-bar rotors.

To summarize the salient features discussed so far,

1. The stator of the 3 - phase induction machine consists of normal distributed AC wind-ings.
2. Balanced three phase voltages impressed on the stator, cause balanced three phase currents to flow in the stator.
3. These stator currents cause a rotating flux pattern (the pattern is a flux distribution which is sinusoidal with respect to the space angle) in the air gap.
4. The rotating flux pattern causes three phase induced e.m.f.s in rotor windings (again normal ac windings). These windings, if shorted, carry three phase-balanced currents. Torque is produced as a result of interaction of the currents and the air gap flux.
5. The rotor may also take the form of a squirrel cage arrangement, which behaves in a manner similar to the short-circuited three phase windings.

Equivalent Circuit

It is often required to make quantitative predictions about the behavior of the induction machine, under various operating conditions. For this purpose, it is convenient to represent the machine as an equivalent circuit under sinusoidal steady state operating conditions. Since the operation is balanced, a single-phase equivalent circuit is sufficient for most purposes.

In order to derive the equivalent circuit, let us consider a machine with an open circuited rotor. Since no current can flow and as a consequence no torque can be produced, the situation is like a transformer open-circuited on the secondary (rotor). The equivalent circuit under this condition can be drawn as shown in fig. 16.

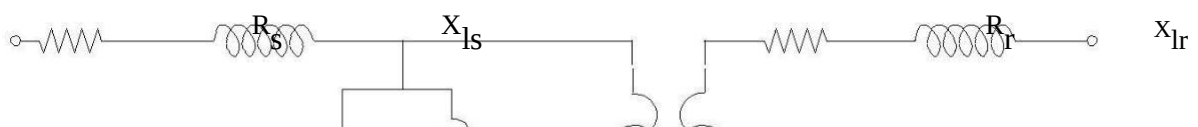


Figure 16: Induction machine with the rotor open

This is just the normal transformer equivalent circuit (why?). Measurements are generally made on the stator side and the rotor, in most circumstances, is shorted (if required, through some external circuitry). Since most of the electrical interaction is from the stator, it makes sense to refer all parameters to the stator.

Let us consider the rotor to be shorted. Let the steady speed attained by the rotor be ω_r and the synchronous speed be ω_s . The induced voltage on the rotor is now proportional to the slip i.e., slip times the induced voltage under open circuit (why?). Further, the voltage induced and the current that flows in the rotor is at a frequency equal to slip times the stator excitation frequency (why?). The equivalent circuit can be made to represent this by shorting the secondary side and is shown in fig. 17.

R'_r and X'_{lr} refer to the rotor resistance and leakage resistance referred to the stator side (using the square of the turns ratio, as is done in transformer). The secondary side loop is excited by a voltage sE_1 , which is also at a frequency sf_1 . This is the reason why the rotor

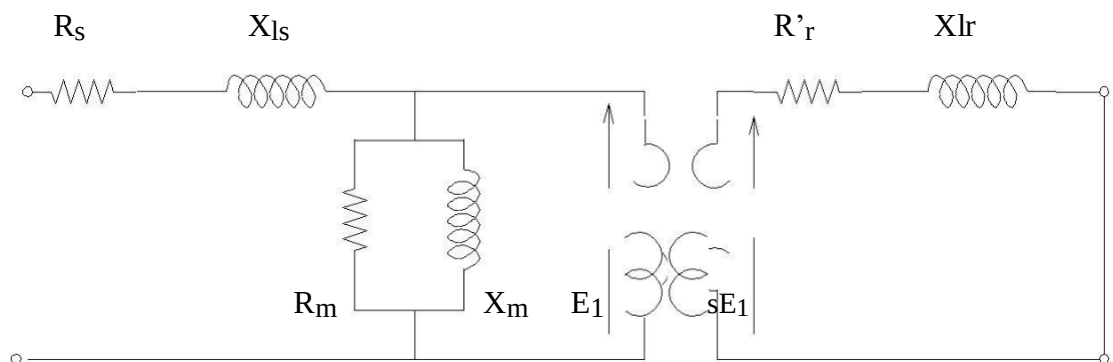


Figure 17: Equivalent circuit : rotor at its own frequency

leakage is sX'_{lr} now . The current amplitude in the rotor side would therefore be

This expression can be modified as follows (dividing numerator and denominator by s)

Equation 7 tells us that the rotor current is the same as the current flowing in a circuit with a load impedance consisting of a resistance R_r' / s and inductive reactance X_{lr}' . This current would also now be at the frequency of E_1 (stator frequency). Note that the slip no longer multiplies the leakage reactance. Further this current is now caused by a voltage of E_1 itself (no multiplying factor of s). Hence the transformer in fig. 17 can also be removed.

Since, with this, the conversion to slip frequency is no longer there, the equivalent circuit can be represented as in fig. 18.

This is then the per-phase equivalent circuit of the induction machine, also called as exact equivalent circuit. Note that the voltage coming across the magnetizing branch is the applied stator voltage, reduced by the stator impedance drop. Generally the stator impedance drop is only a small fraction of the applied voltage. This fact is taken to advantage and the magnetizing branch is shifted to be directly across the input terminals and is shown in fig. 19.

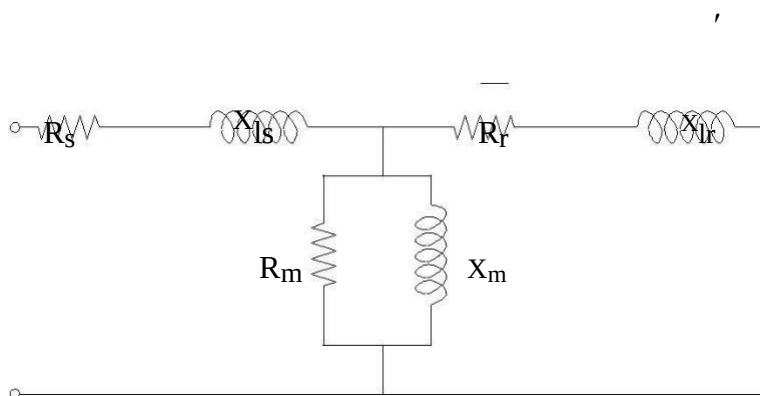


Figure 18: The Exact equivalent circuit

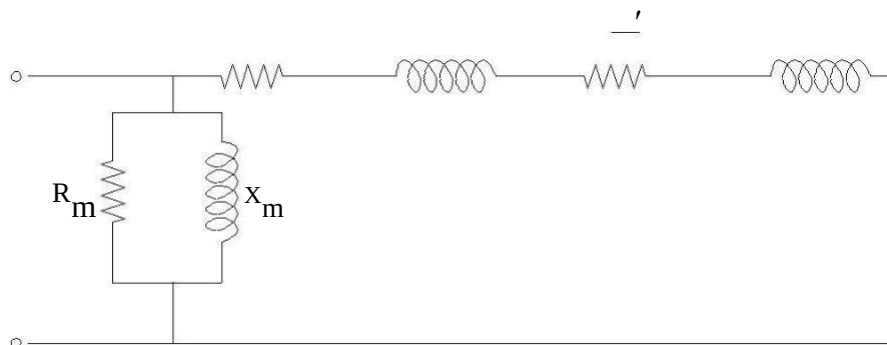


Figure 19: The approximate equivalent circuit

This circuit, called the approximate equivalent circuit, is simple to use for quick calculations. With this equation the equivalent circuit can be modified as shown in fig. 20.

Dividing the equation for the rotor current by s and merging the two sides of the transformer is not just a mathematical jugglery. The power dissipated in the rotor resistance (per phase) is obviously $I_2'^2 R_r'$. From the equivalent circuit of fig. 20 one can see that the rotor current (referred to stator of course) flows through a resistance R_r'/s which has a component $R_r'(1-s)/s$ in addition to R_r' , which also dissipates power. What does this represent?

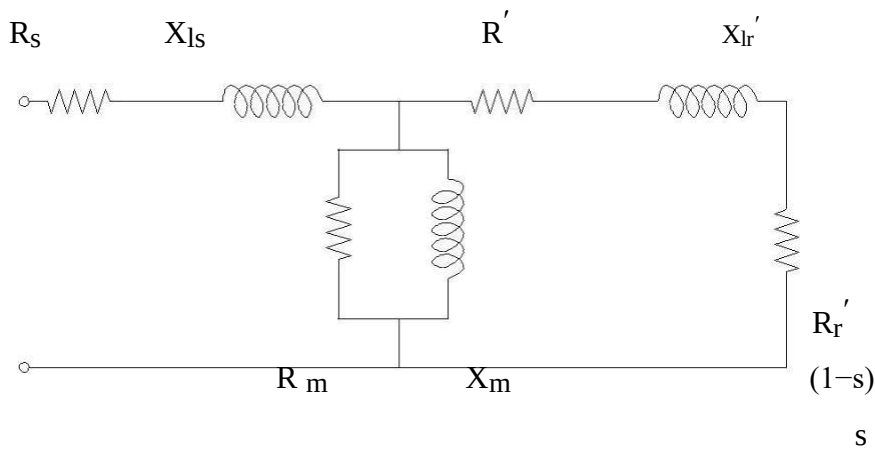


Figure 20: The exact equivalent circuit - separation of rotor resistance

From the equivalent circuit, one can see that the dissipation in R_s represents the stator loss, and dissipation in R_m represents the iron loss. Therefore, the power absorption indicated by the rotor part of the circuit must represent all other means of power consumption - the actual mechanical output, friction and windage loss components and the rotor copper loss components. Since the dissipation in R_r' is rotor copper loss, the power dissipation in $R_r'(1-s)/s$ is the sum total of the remaining. In standard terminology, dissipation in

- R_r'/s is called the air gap power.
- R_r' is the rotor copper loss.

- $R_r' (1 - s)/s$ is the mechanical output.

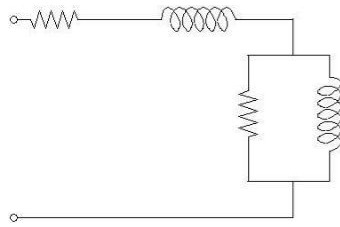
In an ideal case where there are no mechanical losses, the last term would represent the actual output available at the shaft. Out of the power P_g Transferred at the air gap, a fraction s is dissipated in the rotor and $(1 - s)$ is delivered as output at the shaft. If there are no mechanical losses like friction and windage, this represents the power available to the load.

Determination of Circuit Parameters

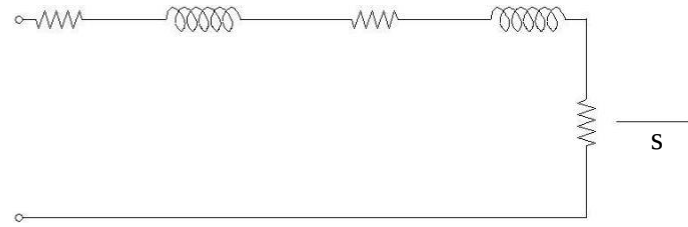
In order to find values for the various elements of the equivalent circuit, tests must be conducted on a particular machine, which is to be represented by the equivalent circuit. In order to do this, we note the following.

1. When the machine is run on no-load, there is very little torque developed by it. In an ideal case where there is no mechanical losses, there is no mechanical power developed at no-load. Recalling the explanations in the section on torque production, the flow of current in the rotor is indicative of the torque that is produced. If no torque is produced, one may conclude that no current would be flowing in the rotor either. The rotor branch acts like an open circuit. This conclusion may also be reached by reasoning that when there is no load, an ideal machine will run up to its synchronous speed where the slip is zero resulting in an infinite impedance in the rotor branch.
2. When the machine is prevented from rotation, and supply is given, the slip remains at unity. The elements representing the magnetizing branch R_m & X_m are high impedances much larger than R_r' & X_{lr}' in series. Thus, in the exact equivalent circuit of the induction machine, the magnetizing branch may be neglected.

From these considerations, we may reduce the induction machine exact equivalent circuit of fig.18 to those shown in fig. 21



(a) No-load equivalent circuit



(b) Blocked rotor equivalent circuit

Figure 21: Reduced equivalent circuits

These two observations and the reduced equivalent circuits are used as the basis for the two most commonly used tests to find out the equivalent circuit parameters — the blocked rotor test and no load test. They are also referred to as the short circuit test and open circuit test respectively in conceptual analogy to the transformer.

The no-load test

The behaviour of the machine may be judged from the equivalent circuit of fig. 21(a). The current drawn by the machine causes a stator-impedance drop and the balance voltage is applied across the magnetizing branch. However, since the magnetizing branch impedance is large, the current drawn is small and hence the stator impedance drop is small compared to the applied voltage (rated value). This drop and the power dissipated in the stator resistance are therefore neglected and the total power drawn is assumed to be consumed entirely as core loss. This can also be seen from the approximate equivalent circuit, the use of which is justified by the foregoing arguments. This test therefore enables us to compute the resistance and inductance of the magnetizing branch in the following manner.

Let applied voltage = V_s . Then current drawn is given by

$$I_s = \frac{V_s}{R_m} + \frac{V_s}{jX_m} \quad (9)$$

The power drawn is given by

$$P_s = \frac{V_s^2}{R_m} \Rightarrow R_m = \frac{V_s^2}{P_s} \quad (10)$$

V_s , I_s and P_s are measured with appropriate meters. With R_m known from eqn. 10, X_m can be found from eqn. 9. The current drawn is at low power factor and hence a suitable wattmeter should be used.

Blocked-rotor Test

In this test the rotor is prevented from rotation by mechanical means and hence the name. Since there is no rotation, slip of operation is unity, $s = 1$. The equivalent circuit valid under these conditions is shown in fig. 21(b). Since the current drawn is decided by the resistance and leakage impedances alone, the magnitude can be very high when rated voltage is applied. Therefore in this test, only small voltages are applied — just enough to cause rated current to flow. While the current magnitude depends on the resistance and the reactance, the power drawn depends on the resistances.

The parameters may then be determined as follows. The source current and power drawn may be written as

$$I_s = \frac{V_s}{(R_s + R'_r) + j(X_s + X'_r)} \quad (11)$$

$$P_s = |I_s|^2 (R_s + R'_r) \quad (12)$$

In the test V_s , I_s and P_s are measured with appropriate meters. Equation 12 enables us to compute $(R_s + R'_r)$. Once this is known, $(X_s + X'_r)$ may be computed from the eqn. 11.

Note that this test only enables us to determine the series combination of the resistance and the reactance only and not the individual values. Generally, the individual values are assumed to be equal; the assumption $R_s = R'_r$, and $X_s = X'_r$ suffices for most purposes. In practice, there are

differences. If more accurate estimates are required IEEE guidelines may be followed which depend on the size of the machine.

Note that these two tests determine the equivalent circuit parameters in a 'Stator-referred' sense, i.e., the rotor resistance and leakage inductance are not the actual values but what they 'appear to be' when looked at from the stator. This is sufficient for most purposes as interconnections to the external world are generally done at the stator terminals.

Deducing the machine performance

From the equivalent circuit, many aspects of the steady state behavior of the machine can be deduced. We will begin by looking at the speed-torque characteristic of the machine. We will consider the approximate equivalent circuit of the machine. We have reasoned earlier that the power consumed by the 'rotor-portion' of the equivalent circuit is the power transferred across the air-gap. Out of that quantity the amount dissipated in R_r' is the rotor copper loss and the quantity consumed by $R_r' (1 - s)/s$ is the mechanical power developed. Neglecting mechanical losses, this is the power available at the shaft. The torque available can be obtained by dividing this number by the shaft speed.

The complete torque-speed characteristics

In order to estimate the speed torque characteristic let us suppose that a sinusoidal voltage is impressed on the machine. Recalling that the equivalent circuit is the per-

$$I_s = \frac{V_s}{\frac{R_r}{s} + j(X_{ls} + X_{lr}')} \quad (14)$$

where V_s is the phase voltage phasor and I_s is the current phasor. The magnetizing phase representation of the machine, the current drawn by the circuit is given by

current is neglected. Since this current is flowing through R_r' , the air-gap power is given by The mechanical power output was shown to be $(1 - s)P_g$ (power dissipated in R_r' / s).

The torque is obtained by dividing this by the shaft speed ω_m . Thus we have,

$$\frac{P_g (1 - s)}{\omega_m} = \frac{P_g (1 - s)}{\omega_s (1 - s)} = |I_s|^2 \frac{R_r'}{s \omega_s} \quad (16)$$

where ω_s is the synchronous speed in radians per second and s is the slip. Further, this is the torque produced per phase. Hence the overall torque is given by

$$T_e = \frac{3}{\omega_s} \cdot \frac{V_s^2}{\left(R_s + \frac{R_r'}{s}\right)^2 + \left(X_{ls} + X_{lr}'\right)^2} \cdot \frac{R_r'}{s}$$

The torque may be plotted as a function of 's' and is called the torque-slip (or torque-speed, since slip indicates speed) characteristic — a very important characteristic of the induction machine. Eqn. 17 is valid for a two-pole (one pole pair) machine. In general, this expression should be multiplied by p , the number of pole-pairs. A typical torque-speed characteristic is shown in fig. 22. This plot corresponds to a 3 kW, 4 pole, 60 Hz machine. The rated operating speed is 1780 rpm.

We must note that the approximate equivalent circuit was used in deriving this relation. Readers with access to MATLAB or suitable equivalents (octave, scilab available free under GNU at the time of this writing) may find out the difference caused by using the 'exact' equivalent circuit by using the script found here. A comparison between the two is found in the plot of fig. 23. The plots correspond to a 3 kW, 4 pole, 50 Hz machine, with a rated speed of 1440 rpm. It can be seen that the approximate equivalent circuit is a good approximation in the operating speed range of the machine. Comparing fig. 22 with fig. 23, we can see that the slope and shape of the characteristics are dependent intimately on the machine parameters.

Further, this curve is obtained by varying slip with the applied voltage being held constant. Coupled with the fact that this is an equivalent circuit valid under steady state, it implies that if this characteristic is to be measured experimentally, we need to look at the torque for a given speed after all transients have died down. One cannot, for example, try to obtain this curve by directly starting the motor with full voltage applied to the terminals

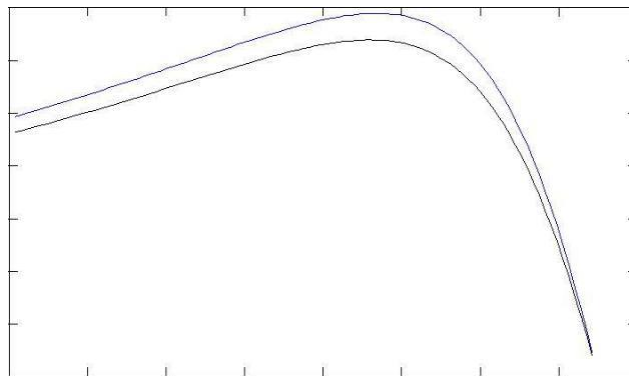


Figure 23: Comparison of exact and approximate circuit predictions

and measuring the torque and speed dynamically as it runs up to steady speed. Another point to note is that the equivalent circuit and the values of torque predicted is

valid when the applied voltage waveform is sinusoidal. With non-sinusoidal voltage wave-forms, the procedure is not as straightforward.

With respect to the direction of rotation of the air-gap flux, the rotor maybe driven to higher speeds by a prime mover or may also be rotated in the reverse direction. The torque-speed relation for the machine under the entire speed range is called the complete speed-torque characteristic. A typical curve is shown in fig. 7.1 for a four-pole machine, the synchronous speed being 1500 rpm. Note that negative speeds correspond to slip values greater than 1, and speeds greater than 1500 rpm correspond to negative slip. The plot also shows the operating modes of the induction machine in various regions. The slip axis is also shown for convenience.

Restricting ourselves to positive values of slip, we see that the curve has a peak point. This is the maximum torque that the machine can produce, and is called as stalling torque. If the load torque is more than this value, the machine stops rotating or stalls. It occurs at a slip $s^{\wedge}$, which for the machine of fig. 7.1 is 0.38. At values of slip lower than $s^{\wedge}$, the curve falls steeply down to zero at $s = 0$. The torque at synchronous speed is therefore zero. At values of slip higher than $s = s^{\wedge}$, the curve falls slowly to a minimum value at $s = 1$. The torque at $s = 1$ (speed = 0) is called the starting torque.

The value of the stalling torque may be obtained by differentiating the expression for torque with respect to zero and setting it to zero to find the value of $s^{\wedge}$. Using this method

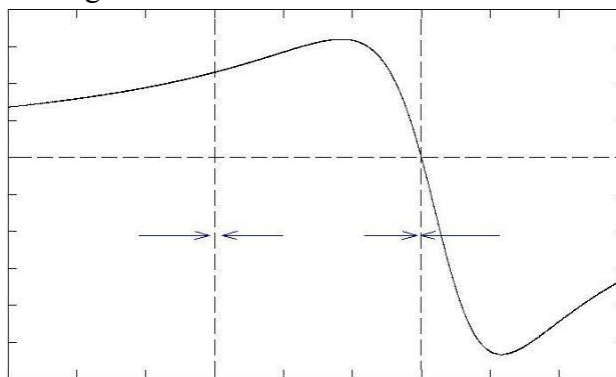


Figure 24: Complete speed-torque characteristic

This fact can be made use of conveniently to alter $s^{\wedge}$. If it is possible to change R_r' , then we can get a whole series of torque-speed characteristics, the maximum torque

remaining constant all the while. But this is a subject to be discussed later.

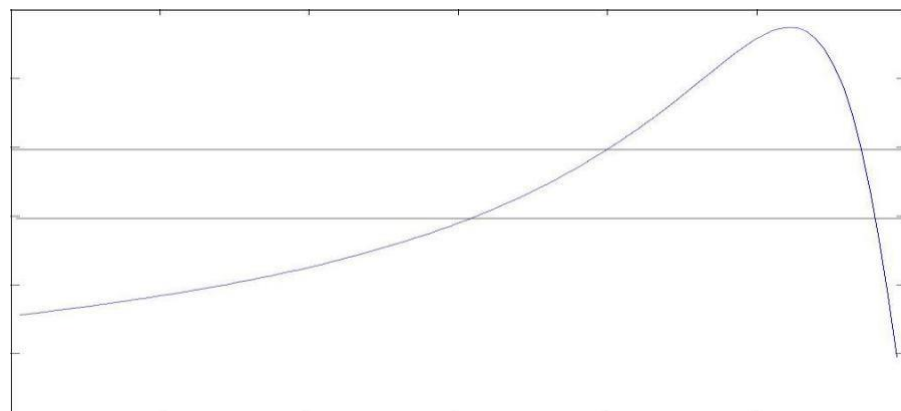
While considering the negative slip range, (generator mode) we note that the maximum torque is higher than in the positive slip region (motoring mode)

Operating Point

Consider a speed torque characteristic shown in fig. 25 for an induction machine, having the load characteristic also superimposed on it. The load is a constant torque load i.e., the torque required for operation is fixed irrespective of speed.

The system consisting of the motor and load will operate at a point where the two characteristics meet. From the above plot, we note that there are two such points. We therefore need to find out which of these is the actual operating point.

To answer this we must note that, in practice, the characteristics are never fixed; they change slightly with time. It would be appropriate to consider a small band around the curve drawn where the actual points of the characteristic will lie. This being the case let us consider that the system is operating at point 1, and the load torque demand increases slightly. This is shown in fig. 26, where the change is exaggerated for clarity. This would shift the point of operation to a point 1' at which the slip would be less and the developed torque higher.



speed, rpm

Figure 26: Stability of operating point

The difference in torque developed ΔT_e , being positive will accelerate the machine.

Any overshoot in speed as it approaches the point $1'$ will cause it to further accelerate since the developed torque is increasing. Similar arguments may be used to show that if for some reason the developed torque becomes smaller the speed would drop and the effect is cumulative. Therefore we may conclude that 1 is not a stable operating point.

Let us consider the point 2. If this point shifts to $2'$, the slip is now higher (speed is lower) and the positive difference in torque will accelerate the machine. This behavior will tend to bring the operating point towards 2 once again. In other words, disturbances at point 2 will not cause a runaway effect. Similar arguments may be given for the case where the load characteristic shifts down. Therefore we conclude that point 2 is a stable operating point.

From the foregoing discussions, we can say that the entire region of the speed-torque characteristic from $s = 0$ to $s = s^{\wedge}$ is an unstable region, while the region from $s = s^{\wedge}$ to $s = 0$ is a stable region. Therefore the machine will always operate between $s = 0$ and $s = s^{\wedge}$

Modes of Operation

The reader is referred to fig. 7.1 which shows the complete speed-torque characteristic of the induction machine along with the various regions of operation.

Let us consider a situation where the machine has just been excited with three phase supply and the rotor has not yet started moving. A little reflection on the definition of the slip indicates that we are at the point $s = 1$. When the rotating magnetic field is set up due to stator currents, it is the induced emf that causes current in the rotor, and the interaction between the two causes torque. It has already been pointed out that it is the

presence of the non-zero slip that causes a torque to be developed. Thus the region of the curve between $s = 0$ and $s = 1$ is the region where the machine produces torque to rotate a passive load and hence is called the motoring region. Note further that the direction of rotation of the rotor is the same as that of the air gap flux.

Suppose when the rotor is rotating, we change the phase sequence of excitation to the machine. This would cause the rotating stator field to reverse its direction — the rotating stator mmf and the rotor are now moving in opposite directions. If we adopt the convention that positive direction is the direction of the air gap flux, the rotor speed would then be a negative quantity. The slip would be a number greater than unity. Further, the rotor as we know should be "dragged along" by the stator field. Since the rotor is rotating in the opposite direction to that of the field, it would now tend to slow down, and reach zero speed. Therefore this region ($s > 1$) is called the

braking region. (What would happen if the supply is not cut-off when the speed reaches zero?)

There is yet another situation. Consider a situation where the induction machine is operating from mains and is driving an active load (a load capable of producing rotation by itself). A typical example is that of a windmill, where the fan like blades of the wind mill are connected to the shaft of the induction machine. Rotation of the blades may be caused by the motoring action of the machine, or by wind blowing. Further suppose that both acting independently cause rotation in the same direction. Now when both grid and wind act, a strong wind may cause the rotor to rotate faster than the mmf produced by the stator excitation. A little reflection shows that slip is then negative. Further, the wind is rotating the rotor to a speed higher than what the electrical supply alone would cause. In order to do this it has to contend with an opposing torque generated by the machine preventing the speed build up. The torque generated is therefore negative. It is this action of the wind against the torque of the machine that enables wind-energy generation. The region of slip $s > 1$ is the generating mode of operation. Indeed this is at present the most commonly used approach in wind-energy generation. It may be noted from the torque expression of eqn. 17 that torque is negative for negative values of slip.

Speed control of Induction Machines

We have seen the speed torque characteristic of the machine. In the stable region of operation in

the motoring mode, the curve is rather steep and goes from zero torque at synchronous speed to the stall torque at a value of slip $s = s^{\wedge}$. Normally $s^{\wedge}$ may be such that stall torque is about three times that of the rated operating torque of the machine, and hence may be about 0.3 or less. This means that in the entire loading range of the machine, the speed change is quite small. The machine speed is quite stiff with respect to load changes. The entire speed variation is only in the range n_s to $(1 - s^{\wedge})n_s$, n_s being dependent on supply frequency and number of poles.

The foregoing discussion shows that the induction machine, when operating from mains is essentially a constant speed machine. Many industrial drives, typically for fan or pump applications, have typically constant speed requirements and hence the induction machine is ideally suited for these. However, the induction machine, especially the squirrel cage type, is quite rugged and has a simple construction. Therefore it is good candidate for variable speed applications if it can be achieved.

UNIT-V**Speed control****Speed control by changing applied voltage**

From the torque equation of the induction machine given in eqn.17, we can see that the torque depends on the square of the applied voltage. The variation of speed torque curves with respect to the applied voltage is shown in fig. 27. These curves show that the slip at maximum torque $s^{\wedge}$ remains same, while the value of stall torque comes down with decrease in applied voltage. The speed range for stable operation remains the same.

Further, we also note that the starting torque is also lower at lower voltages. Thus, even if a given voltage level is sufficient for achieving the running torque, the machine may not start. This method of trying to control the speed is best suited for loads that require very little starting torque, but their torque requirement may increase with speed.

Figure 27 also shows a load torque characteristic — one that is typical of a fan type of load. In a fan (blower) type of load, the variation of torque with speed is such that $T \propto \omega^2$. Here one can see that it may be possible to run the motor to lower speeds within the range n_s to $(1 - s^{\wedge})n_s$. Further, since the load torque at zero speed is zero, the machine can start even at reduced voltages. This will not be possible with constant torque type of loads.

One may note that if the applied voltage is reduced, the voltage across the magnetising branch also comes down. This in turn means that the magnetizing current and hence flux level are reduced. Reduction in the flux level in the machine impairs torque production

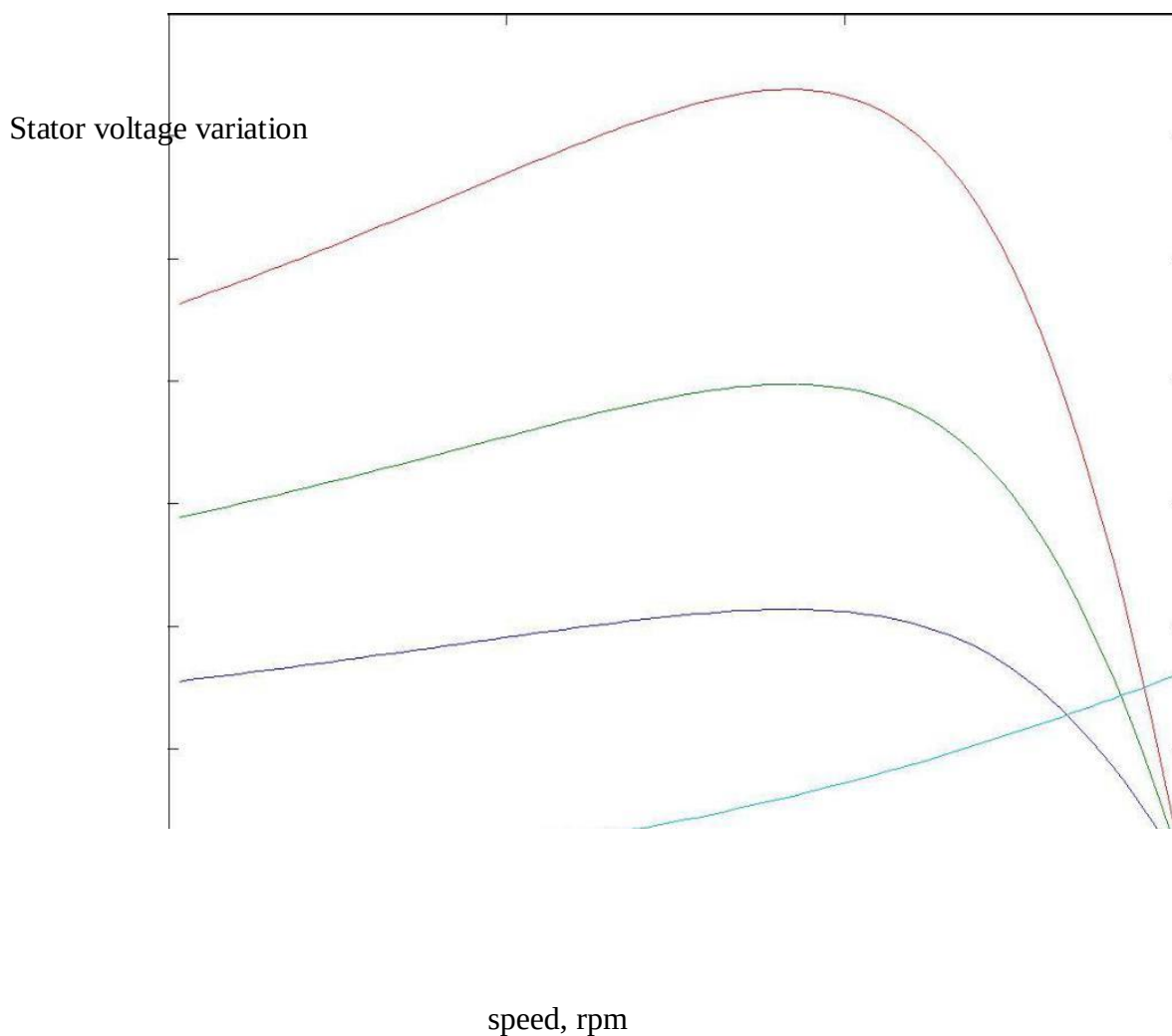


Figure 27: Speed-torque curves: voltage variation

(recall explanations on torque production), which is primarily the explanation for fig. 27. If, however, the machine is running under lightly loaded conditions, then operating under rated flux levels is not required. Under such conditions, reduction in magnetizing current improves the power factor of operation. Some amount of energy saving may also be achieved.

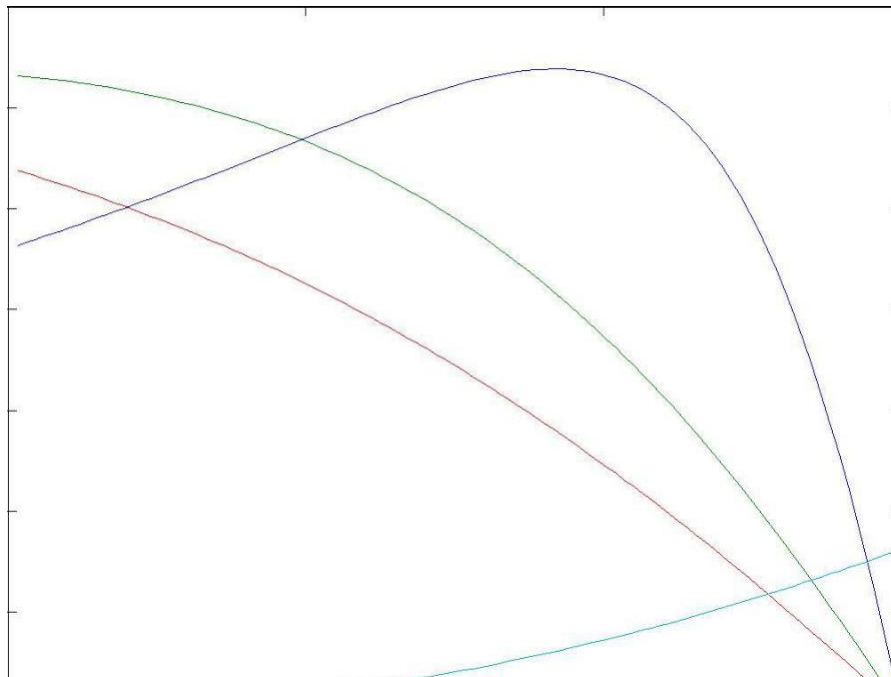
Voltage control may be achieved by adding series resistors (a lossy, inefficient proposition), or a series inductor / autotransformer (a bulky solution) or a more modern solution using semiconductor devices. A typical solid state circuit used for this purpose is the AC voltage controller or AC chopper. Another use of voltage control is in the so-called 'soft-start' of the machine. This is discussed in the section on starting methods.

Rotor resistance control

The reader may recall from eqn.17 the expression for the torque of the induction machine. Clearly, it is dependent on the rotor resistance. Further, eqn.19 shows that the maximum value is independent of the rotor resistance. The slip at maximum torque eqn.18 is dependent on the rotor

resistance. Therefore, we may expect that if the rotor resistance is changed, the maximum torque point shifts to higher slip values, while retaining a constant torque. Figure 28 shows a family of torque-speed characteristic obtained by changing the rotor resistance.

Rotor resistance variation



speed, rpm

Figure 28: Speed-torque curves : rotor resistance variation

Note that while the maximum torque and synchronous speed remain constant, the slip at which maximum torque occurs increases with increase in rotor resistance, and so does the starting torque. whether the load is of constant torque type or fan-type, it is evident that the speed control range is more with this method. Further, rotor resistance control could also be used as a means of generating high starting torque.

For all its advantages, the scheme has two serious drawbacks. Firstly, in order to vary the rotor resistance, it is necessary to connect external variable resistors (winding resistance itself cannot be changed). This, therefore necessitates a slip-ring machine, since only in that case rotor terminals are available outside. For cage rotor machines, there are no rotor terminals. Secondly, the method is not very efficient since the additional resistance and operation at high slips entails dissipation.

The resistors connected to the slip-ring brushes should have good power dissipation capability. Water based rheostats may be used for this. A 'solid-state' alternative to a rheostat is a chopper controlled resistance where the duty ratio control of the chopper presents a variable resistance load to the rotor of the induction machine.

Cascade control

The power drawn from the rotor terminals could be spent more usefully. Apart from using the heat generated in meaningful ways, the slip ring output could be connected to another induction machine. The stator of the second machine would carry slip frequency currents of the first machine which would generate some useful mechanical power. A still better option would be to mechanically couple the shafts of the two machines together. This sort of a connection is called cascade connection and it gives some measure of speed control as shown below.

Let the frequency of supply given to the first machine be f_1 , its number poles be p_1 , and its slip of operation be s_1 . Let f_2 , p_2 and s_2 be the corresponding quantities for the second machine. The frequency of currents flowing in the rotor of the first machine and hence in the stator of the second machine is $s_1 f_1$. Therefore $f_2 = s_1 f_1$. Since the machines are coupled at the shaft, the speed of the rotor is common for both. Hence, if n is the speed of the rotor

per phase

$$sE_1 I_2' \cos \phi_2 = I_2'^2 R_2' + P_r . \quad (22)$$

Clearly now, the value of s can be changed by the value of P_r . For $P_r = 0$, the machine is like a normal machine with a short circuited rotor. As P_r becomes positive, for all other circuit conditions remaining constant, s increases or in the other words, speed reduces. As P_r becomes negative, the right hand side of the equation and hence the slip decreases. The physical interpretation is that we now have an active source connected on the rotor side which is able to supply part of the rotor copper losses. When $P_r = -I_2'^2 R_2'$ the entire copper loss is supplied by the external source. The RHS and hence the slip is zero. This corresponds to operation at synchronous speed. In general the circuitry connected to the rotor may not be a simple resistor or a machine but a power electronic circuit which can process this power requirement. This circuit may drive a machine or recover power back to the mains. Such circuits are called static kramer drives.

Pole changing schemes

Sometimes induction machines have a special stator winding capable of being externally connected to form two different number of pole numbers. Since the synchronous speed of the induction machine is given by $n_s = f_s/p$ (in rev./s) where p is the number of pole pairs, this would correspond to changing the synchronous speed. With the slip now corresponding to the new synchronous speed, the operating speed is changed. This method of speed control is a stepped variation and generally restricted to two steps.

If the changes in stator winding connections are made so that the air gap flux remains constant, then at any winding connection, the same maximum torque is achievable. Such winding arrangements are therefore referred to as constant-torque connections. If however such connection changes result in air gap flux changes that are inversely proportional to the synchronous speeds, then such connections are called constant-horsepower type.

The following figure serves to illustrate the basic principle. Consider a magnetic pole structure consisting of four pole faces A, B, C, D as shown in fig. 30.

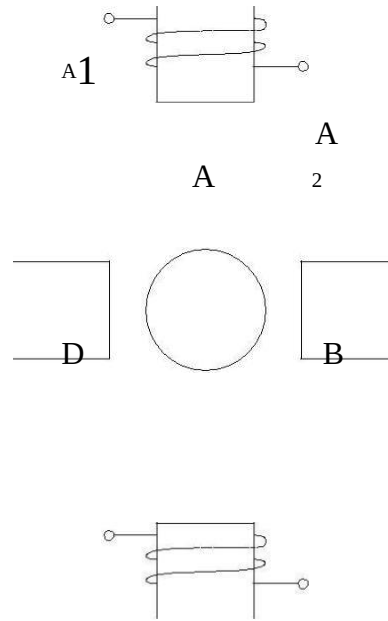


Figure 30: Pole arrangement

Harmonics in Induction Machines

In attempting to understand the performance of an induction machine, we consider that the air-gap flux wave is purely sinusoidal. It is from that assumption the analysis of induced emf, sinusoidal currents, the expressions for generated torque etc. proceed. In practice, there are deviations from this idealistic picture.

Time Harmonic

The first non-ideality is the presence of harmonics in the input supply given to the three phase machine. The source may contain 3rd, 5th, 7th... harmonics. Note that due to the symmetry of the waveform ($f(t) = -f(t + T/2)$), where T is the period of the supply sine waveform, even ordered harmonics cannot exist. Let the R phase supply voltage be given by the expression

$$v_R = V_{1m} \sin(\omega_1 t + \phi_1) + V_{3m} \sin(3\omega_1 t + \phi_3) + V_{5m} \sin(5\omega_1 t + \phi_5) + V_{7m} \sin(7\omega_1 t + \phi_7) + \dots \quad (25)$$

Being a balanced three phase supply, we know that the waveforms of v_Y and v_B are 120° and 240°

Thus the expressions for v_Y and v_B would be

$$\begin{aligned}
 v_Y &= V_{1m} \sin(\omega_1 t + \phi_1 - \frac{2\pi}{3}) + V_{3m} \sin(3\omega_1 t + \phi_3 - 3 \cdot \frac{2\pi}{3}) \\
 &\quad + V_{5m} \sin(5\omega_1 t + \phi_5 - 5 \cdot \frac{2\pi}{3}) + V_{7m} \sin(7\omega_1 t + \phi_7 - 7 \cdot \frac{2\pi}{3}) + \dots \quad (26) \\
 v_B &= V_{1m} \sin(\omega_1 t + \phi_1 - \frac{4\pi}{3}) + V_{3m} \sin(3\omega_1 t + \phi_3 - 3 \cdot \frac{4\pi}{3}) \\
 &\quad + V_{5m} \sin(5\omega_1 t + \phi_5 - 5 \cdot \frac{4\pi}{3}) + V_{7m} \sin(7\omega_1 t + \phi_7 - 7 \cdot \frac{4\pi}{3}) + \dots
 \end{aligned}$$

shifted from v_R respectively. It is further well known that if a waveform is shifted by ϕ degrees, its harmonics are shifted by $n\phi$ degrees, where n is the order of the harmonic.

If we consider the third harmonic components of the three phase waveforms, and if $v_{x3}(t)$ is the third harmonic of phase x , we can see that

$$\begin{aligned}
 v_R &= V_{3m} \sin(3\omega_1 t + \phi_3) \\
 v_Y &= V_{3m} \sin(3\omega_1 t + \phi_3) \\
 v_B &= V_{3m} \sin(3\omega_1 t + \phi_3) \quad (28)
 \end{aligned}$$

Therefore, all the three third harmonics are in phase. In a STAR connected system with isolated neutral, these voltages cannot cause any current flow since all three terminals are equal in potential. If the neutral point is connected to some point, then then current can flow through the neutral connection. Such a connection is however rare in induction machines. The machine is therefore an open circuit to third harmonics. In fact, one can see that any harmonic whose order is a multiple of three, i.e., the triplen harmonics, as they are called, will face an identical situation. Since the machine is an open circuit to triplen harmonics in the excitation voltage, these do not have effect on the machine.

Let us now consider the fifth harmonic. From the equations above, one can see that

$$\begin{aligned}
 v_{RS} &= V_{5m} \sin(5\omega_1 t + \phi_5) \\
 v_Y &= V_{5m} \sin\left(5\omega_1 t + \phi_5 - \frac{5 \cdot 3}{4\pi}\right) \\
 &= V_{5m} \sin\left(5\omega_1 t + \phi_5 - \frac{3}{4\pi}\right) \\
 v_{BS} &= V_{5m} \sin\left(5\omega_1 t + \phi_5 - \frac{5 \cdot 3}{2\pi}\right) \\
 &= V_{5m} \sin\left(5\omega_1 t + \phi_5 - \frac{3}{2}\right) \quad (29)
 \end{aligned}$$